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The *UBE3A*-ATS antisense oligonucleotide rugonersen in children with Angelman syndrome: a phase 1 trial

In the format provided by the authors and unedited

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1. Clinical sites, principal investigators and IRB / ethic committee IDs

Site	PI	Country	EC / IRB	EC / IRB Identifier
Ospedale Pediatrico Bambino Gesu	Dr Pietrafusa (Current) Dr Vigevano (Past)	Rome, Italy	Comitato Etico dell' IRCCS Ospedale Pediatrico Bambino Gesù	2084 / 2020
Erasmus MC - Erasmus Medisch Centrum	Dr de Wit	Rotterdam, Netherlands	Central Committee on Research Involving Human Subjects (CCMO)	NL73321.000.20
H. Sant Joan de Deu, Barcelona	Dr Serrano Gimare	Barcelona, Spain	CEIC Fundació Sant Joan de Dewu, Servei de Recerca	AC-03-20
Parc Taulí, Sabadell	Dr Roche	Barcelona, Spain	CEIC Fundació Sant Joan de Dewu, Servei de Recerca	AC-03-20
Hospital Virgen del Rocio, Sevilla	Dr Muñoz	Sevilla, Spain	CEIC Fundació Sant Joan de Dewu, Servei de Recerca	AC-03-20
RUSH Children's Hospital	Dr Berry-Kravis	Chicago, United States	Rush University Medical Center, Rush University Medical Center Institutional Review Board	00000482
Texas Children's Hospital	Dr Bacino	Houston, United States	Baylor College of Medicine Office of Research, Institutional Review Board	H-47303
UNC School of Medicine - Carolina Institute for Developmental Disabilities	Dr Shen (Current) Dr Capal (Past)	Chapel, Hill, United States	Advarra	00041741
UCLA	Dr Rajamaran (Current) Dr Jeste (Past)	Los Angeles, United States	Advarra	00041741
Rady Children's Hospital	Dr Bird	San Diego, United States	UCSD Human Research Protections Program, Altman Clinical and Translational Research Institute	200683
Columbia University Medical Center	Dr Bain	New York, United States	Columbia University Medical Center, CUMC IRB	AAAS9480
Mayo Clinic, Rochester	Dr Gavrilova	Rochester, United States	Mayo Clinic Institutional Review Board Rochester	20-007499

Table S1. Clinical sites, principal investigators and Institutional Review Board (IRB) / ethic committee (EC) IDs.

2. The Rugonersen study group

João A. Abrantes, Chai Adams, Angelique Augustin, Edward Austin, Muhamed Barakovic, Raman Bedi, Alessandro Belotti, Olga Bettoni, Charlotte Bon, Azad Bonni, Ponghatai Boonsimma, Anna Brändli-Baiocco, Monika Broennimann, Gary Brouwer, Ina Bruenig Traebert, Mariana Bustamante, Katalin Buzasi, Stormy Chamberlain, Daniel Chan, Susanne Clinch, Siobhan Connor-Ahmad, Gianriccardo Corbo, Philipp Daetwyler, Oana Danila, Cassandra Dean, Axel Ducret, Thomas Emrich, Ela Erturk, Sam Ewing, Paulo Fontoura, Isabel Galnares, Hans Peter Grimm, Leo Gschwind, Alexandra Hawes, Jörg F. Hipp, Marius Hoener, Sebastian Holst, Michael Honer, Gregory Hooper, Jörg Hörnschemeyer, Florian Islinger, Ravi Jagasia, Ivan Jelcic, Ellen Ji, Magdalena Jodlowska, Krisztina Kajzinger, Geoff Kerchner, Gregory Klein, Erich Koller, Vitaliy Kolodyazhniy, Thomas Kremer, Michelle L. Krishnan, Gigi Lee, Valentin Legras, Rea Lehner, Jörg Lill, Dominick Lionetti, Florian Lipsmeier, Jon Lopez, Renee Maas, Stefano Magon, Harald Mauser, Eilidh Mayer, Martin Mehlmann, Helene Meistermann, Amanda Messer, Chantal Michaud, Rema Moore, Anna Morka, Lutz Müller, Lorraine Murtagh, David Nobbs, Alessandro Noci, Robyn OConnor, Giuseppe Palermo, Nikhil Pandya, Christophe Pfleger, Emma Poon, Lisa Restelli, Sophie Schaller, Dietmar Schwab, Shady Sedhom, Sushma Shah, Hanna E. Silber Baumann, Kamil Sklodowski, Siegfried Stolz, Lisa Squassante , Julian Tillmann, Jorrit Tjeertes, Manuel Tzouros, Naina Varsani, Brenda Vincenzi, Barbara Vinduska, Franziska Voellmy, David Stephen Ward, Thomas Wiese, Penelope Wilks, Tom Willgoss, Jenna Willis, Saskia Zaaijer

3. Study Participants

Characteristic	A Cohorts (Age 5 – 12 years) A1 (n=6)	A2 (n=8)	A3 (n=7)	A4 (n=5)	EA1 (n=5)	B Cohorts (Age 1 – 4 years) B1 (n=6)	B2 (n=6)	B3 (n=6)	B4 (n=6)	EB1 (n=6)	All (n=61)
FMS TOTAL SCORE (SD)	14.0 (6.7)	14.4 (1.8)	12.9 (7.2)	8.2 (5.6)	5.5 (7.8)	11.3 (6.2)	6.8 (6.4)	9.7 (5.2)	14.6 (3.0)	3.2 (4.8)	10.3 (6.3)
ORCA Total T-Score v2 (SD)	45.0 (2.1)	48.8 (8.6)	50.6 (12.0)	38.5 (6.9)	43.2 (12.4)	42.9 (6.9)	45.1 (4.3)	45.1 (3.9)	48.6 (7.5)	41.6 (3.7)	45.4 (7.8)
Aberrant Behavior Checklist Second Edition – Community Version (ABC-2C), Raw Scores (SD)											
Hyperactivity / Non-Compliance	18.7 (8.7)	18.1 (9.0)	16.4 (10.3)	21.0 (14.1)	19.8 (13.2)	24.0 (6.8)	14.2 (10.3)	13.5 (9.0)	20.4 (11.4)	10.0 (6.6)	17.5 (10.1)
Inappropriate Speech	0.8 (2.0)	1.4 (2.5)	0.7 (1.5)	0.0 (0.0)	0.0 (0.0)	0.8 (1.3)	0.3 (0.5)	0.0 (0.0)	0.2 (0.4)	0.0 (0.0)	0.5 (1.3)
Irritability	9.8 (9.2)	14.6 (8.8)	6.7 (5.0)	9.8 (7.3)	7.2 (7.5)	7.2 (4.6)	5.2 (7.4)	5.3 (6.3)	10.0 (8.5)	4.5 (5.0)	8.2 (7.3)
Social Withdrawal	5.7 (8.5)	7.8 (11.4)	3.9 (3.5)	6.4 (2.2)	6.7 (5.7)	7.3 (6.3)	3.8 (2.6)	2.5 (3.7)	4.0 (2.5)	3.7 (2.7)	5.2 (5.9)
Stereotypic Behavior	4.0 (3.0)	4.2 (2.8)	4.6 (3.5)	8.6 (6.1)	7.3 (6.7)	4.7 (3.4)	4.0 (3.2)	2.3 (1.4)	6.2 (3.5)	2.7 (1.9)	4.8 (3.9)
Schlaffragebogen für Kinder mit Neurologischen und Anderen Komplexen Erkrankungen (SNAKE), T-Scores (SD)											
Disturbances going to sleep	50.0 (10.3)	50.0 (8.6)	50.4 (10.3)	49.6 (3.6)	51.2 (7.3)	55.0 (6.4)	51.5 (2.0)	50.3 (8.4)	52.0 (6.8)	48.7 (8.3)	50.8 (7.4)
Disturbances remaining asleep	49.2 (7.1)	48.6 (6.3)	45.7 (7.6)	52.6 (7.5)	54.5 (9.1)	51.7 (8.5)	51.2 (3.4)	50.8 (9.8)	54.4 (8.2)	55.8 (4.5)	51.2 (7.5)
Arousal disorders	43.7 (5.1)	43.0 (4.6)	45.3 (3.9)	43.4 (2.6)	44.2 (4.3)	46.8 (6.4)	44.0 (4.6)	43.8 (4.5)	45.8 (7.8)	42.2 (3.6)	44.2 (4.7)
Daytime sleepiness	46.0 (4.9)	44.0 (7.3)	40.0 (5.5)	49.6 (8.0)	51.5 (8.5)	45.2 (7.9)	51.7 (3.3)	50.8 (8.5)	42.6 (8.0)	56.3 (6.9)	47.6 (8.1)
Daytime behavior disorders	53.7 (6.8)	51.5 (8.1)	45.3 (5.5)	54.2 (8.2)	53.0 (6.0)	48.0 (7.0)	49.8 (7.7)	51.7 (10.0)	53.6 (9.6)	46.0 (7.3)	50.5 (7.7)
Caregiver-reported Angelman Syndrome Scale for Impact (CASS-I) (SD)											
Impact by Seizures	-	-	1.2 (0.4)	1.2 (0.4)	-	1.3 (0.5)	1.3 (0.5)	2.2 (1.8)	1.6 (0.9)	1.7 (0.8)	1.5 (0.9)
Sleep Throughout the Night	-	-	2.5 (1.6)	2.6 (1.5)	-	2.7 (1.4)	2.7 (1.5)	2.4 (0.9)	3.6 (1.5)	4.2 (0.8)	2.9 (1.4)
Disruptive Behavior	-	-	2.3 (1.0)	2.0 (0.7)	-	2.7 (1.2)	2.7 (1.6)	2.8 (1.8)	3.6 (0.9)	1.5 (0.5)	2.5 (1.3)
Communicate with People	-	-	4.2 (1.2)	4.8 (0.4)	-	4.0 (1.3)	4.5 (1.2)	4.2 (0.8)	4.2 (0.8)	4.7 (0.5)	4.4 (0.9)
Perform Tasks using Hands	-	-	3.7 (1.2)	3.8 (1.3)	-	3.5 (1.2)	3.8 (0.4)	2.2 (1.3)	2.8 (1.1)	3.0 (1.5)	3.3 (1.2)
Walk	-	-	2.0 (1.5)	4.4 (0.9)	-	2.8 (1.5)	4.3 (0.8)	3.2 (2.0)	2.2 (1.1)	4.2 (1.6)	3.3 (1.6)
Concentrate / Follow Instructions	-	-	3.8 (1.0)	4.8 (0.4)	-	3.8 (1.3)	4.5 (0.8)	3.4 (1.5)	4.0 (0.7)	4.5 (0.8)	4.1 (1.0)
Perform Self-Care Activities	-	-	4.3 (0.8)	4.8 (0.4)	-	5.0 (0.0)	4.8 (0.4)	4.6 (0.9)	4.2 (1.3)	5.0 (0.0)	4.7 (0.7)
AS Symptoms Overall	-	-	3.3 (1.2)	4.2 (1.3)	-	3.7 (1.0)	4.2 (0.8)	4.0 (0.7)	4.0 (1.2)	4.7 (0.8)	4.0 (1.0)

Table S2. Participant baseline characteristics for additional clinical scales. *Participant baseline characteristics for additional clinical scales (FMS, ORCA, ABC-2C, SNAKE, CASS-I, see Supplementary Material Section 9 for details) for MAD cohorts B1-B4 and A1-A4, LTE cohorts EA1 and EB1, and the summary across all cohorts (see Table 1 for demographics and baseline characteristics of BSID, VABS and SAS-CGI-S). For the clinical scales, numbers were in part lower than the cohort numbers (line 2) due to few assessments not performed or invalid. The CASS-I was available only following the start of cohorts A1, A2 and B1. SD: standard deviation.*

4. Safety and Tolerability

Adverse events of pyrexia, vomiting, seizure, and ataxia

The most common AEs were pyrexia, vomiting, seizure, and ataxia occurring within 10 days of dose administration (see Table 2). Table S3 provides a listing of frequency, onset and duration by dose.

Dose (mg)+	# Parti-cipants*	# Doses	Pyrexia AEs		Vomiting AEs		Seizure AEs		Ataxia AEs	
			Doses (%)	Onset / Duration (Days)	Doses (%)	Onset / Duration (Days)	Doses (%)	Onset / Duration (Days)	Doses (%)	Onset / Duration (Days)
Cohort A										
60	19	34	4 (11.76)	2.8 / 2.5	8 (23.52)	2.6 / 1.6	2 (5.88)	3.5 / 2.5	1 (2.94)	0.0 / 2.0
90	8	15	9 (60.00)	1.0 / 4.0	7 (46.66)	2.1 / 2.1	0 (0.00)	NA	3 (20.00)	0.3 / 3.0
120	27	120	46 (38.34)	2.5 / 3.4	29 (24.16)	2.6 / 2.2	5 (4.16)	5.4 / 3.2	19 (15.84)	0.2 / 3.7
180	20	55	34 (61.82)	2.2 / 4.3	19 (32.72)	2.9 / 3.3	2 (3.64)	5.0 / 2.0	17 (30.90)	0.3 / 7.9
Cohort B										
60	24	38	6 (15.78)	3.8 / 2.0	2 (5.26)	1.0 / 2.5	3 (7.90)	5.3 / 3.0	3 (7.90)	0.3 / 4.3
90	13	13	6 (46.16)	2.3 / 3.0	3 (23.08)	2.7 / 1.7	0 (0.00)	NA	1 (7.70)	0.0 / 14.0
120	28	93	52 (55.92)	1.9 / 4.0	11 (11.82)	2.8 / 2.2	4 (4.30)	2.6 / 2.8	11 (11.82)	0.5 / 5.0
180	12	39	25 (64.10)	1.4 / 5.0	5 (12.82)	1.4 / 2.0	5 (12.82)	3.7 / 2.5	7 (17.94)	0.1 / 19.6

Table S3. Adverse events of pyrexia, vomiting, seizure, and ataxia occurring within 10 days of dose administration with 60, 90, 120, or 180 mg of rugonersen. *NA: not applicable. +at Scheduled Intervals Ranging from 4 to 24 Weeks. *Participant counts per dose do not sum up to the total of 61 due to multiple doses per participant.*

Laboratory findings in cerebrospinal fluid (CSF)

There were 4 AEs relevant to abnormalities of CSF cells reported in 6 participants. The AEs include CSF white blood cell count increased (3 participants), CSF red blood cell count positive (1 participant), CSF mononuclear cell count increased (1 participant) and CSF test abnormal (verbatim macrophages in CSF; 1 participant). CSF white blood cell count increased in 2 participants, CSF mononuclear count increased and CSF test abnormal (verbatim macrophage in CSF) were deemed related to the study drug. The other AEs were deemed related to LP or other causes. All the AEs were Grade 1. No accompanying neurological changes were reported with the AEs of CSF abnormalities. No treatment was required, the dose was not changed in response to the event and all events resolved. There were no clinically significant changes or trends identified across the cohorts.

Vital signs and ECG

Abnormalities in vital signs were reported, however, there were no clinically significant changes or trends identified across cohorts. One Grade 1 AE of hypotension was reported. The event resolved within 1 day and was considered related to sedation/anesthesia.

Abnormalities in ECG parameters were reported, however, there were no clinically significant changes or trends across the cohorts. Sinus tachycardia was reported in 1 participant on 5 separate occasions (Grade 1 for all episodes), assessed as related to study drug and sedation/anesthesia. All events of sinus tachycardia resolved. Across all cohorts, no high values for QTc according to Fridericia's correction formula were reported, whereas 15 high values were reported according to Bazett's correction formula, with none reported as clinically significant AEs. Confounding factors included administration of concomitant anesthesia. Of note, Frederica's correction formula is recommended for pediatric ECG interpretation[1].

5. Pharmacokinetics

Results

Plasma concentration (see Online Methods) of rugonersen post-dose at day 1 was dose-related, while concentration at later time-points provided limited information. Rugonersen plasma concentrations reached median T_{max} between three- and six hours post-dose (Fig. S1a). Large variability was observed as indicated by geometric coefficient of variation values of up to more than 300%. Measurable plasma concentrations were observed at higher dose levels (≥30 mg) in the MAD at 4 weeks post-dose. Concentrations in pre-dose samples in the LTE (16 weeks or more post dose) were mostly below the level of quantification (114 out of 137 samples).

Rugonersen concentrations in the CSF samples (see Online Methods) poorly reflected the dose levels administered and were of limited value to infer concentration in the target tissue. CSF samples were taken at pre-dose at the dosing events and at different timepoints. Mean CSF concentrations across dose groups showed only a relatively small increase of approximately 2-to-3-fold within a dose range of 15-fold from 6 to 90 mg at samples taken 28 days post dose. Only sparse data were available for 120 and 180 mg as samples got lost during shipment. CSF concentration in pre-dose samples in the LTE (12 weeks or more post dose) were mostly below the level of quantification (>92% samples).

Discussion

Rugonersen plasma or CSF concentrations provided limited information on individual variability of drug effects. Following IT administration, rugonersen peak plasma concentrations were observed within a few hours. Plasma concentration was dose-related but highly variable. The data suggest relatively fast egress of drug from the CNS to plasma. Drug concentration in the CSF was hypothesized to be a better proxy of concentration in the brain compared to plasma concentration. However, there was only a shallow and highly variable relationship between dose and CSF concentration. The data could not be used to infer brain drug concentrations. CSF samples were taken 4 weeks or later following IT administration of rugonersen. CSF samples taken at an earlier point may have produced more informative results. For other ASOs, informative dose-proportional CSF concentrations have been observed[2,3]. We speculate that rugonersen is distributed to brain within a few days, and the CSF drug concentrations observed mainly reflects re-distribution of drug in the CNS, representing probably only a small fraction of rugonersen in brain tissue.

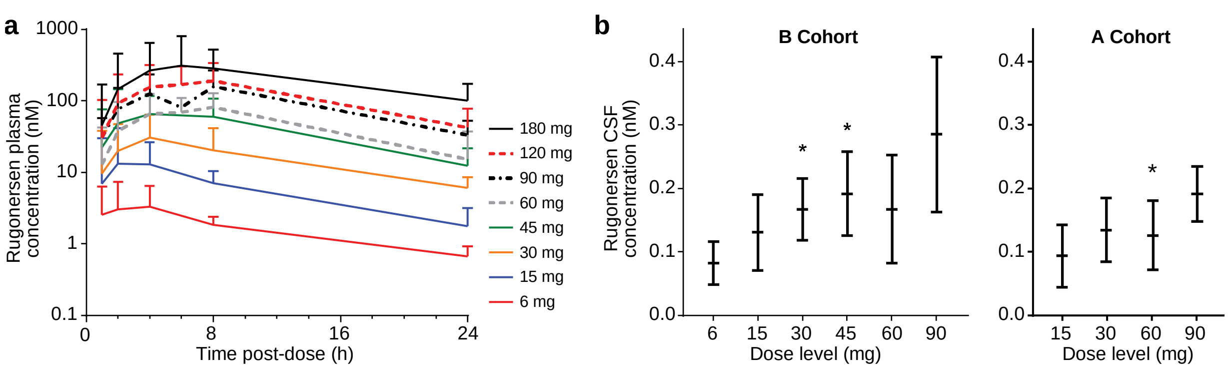

Figure S1. Pharmacokinetics in plasma and CSF. a**, Pharmacokinetics in plasma. Geometric means and geometric coefficients of variation are shown. At doses of 60 mg and higher 6 hours collections were added. All injections at the same dose were pooled. The number of available data per time-point were 5-6, 8-9, 14-17,4- 6, 9-42, 1-21, 24-61, and 15-36 for 6, 15, 30, 45, 60, 90, 120 and 180 mg, respectively. Two* *participants who erroneously received 19.8 mg instead of 15 mg are excluded from this analysis. Data from cut-off date February 2024.* ***b*, Pharmacokinetics in CSF. Mean and standard deviation of rugonersen concentration in CSF at 28 days post last rugonersen dose. Data from MAD weeks 4 and 8 (N=5-8 for A Cohorts and N=6 for B Cohorts). For CSF concentrations below the limit of quantification half of the detection limit was imputed (0.0245 nM). *for each indicated dose group, one value was excluded as outlier (outlier definition: at least 10-fold of the mean without outlier). Data cut-off date: February 2024.*

6. EEG natural history data: developmental trajectory and analytical properties

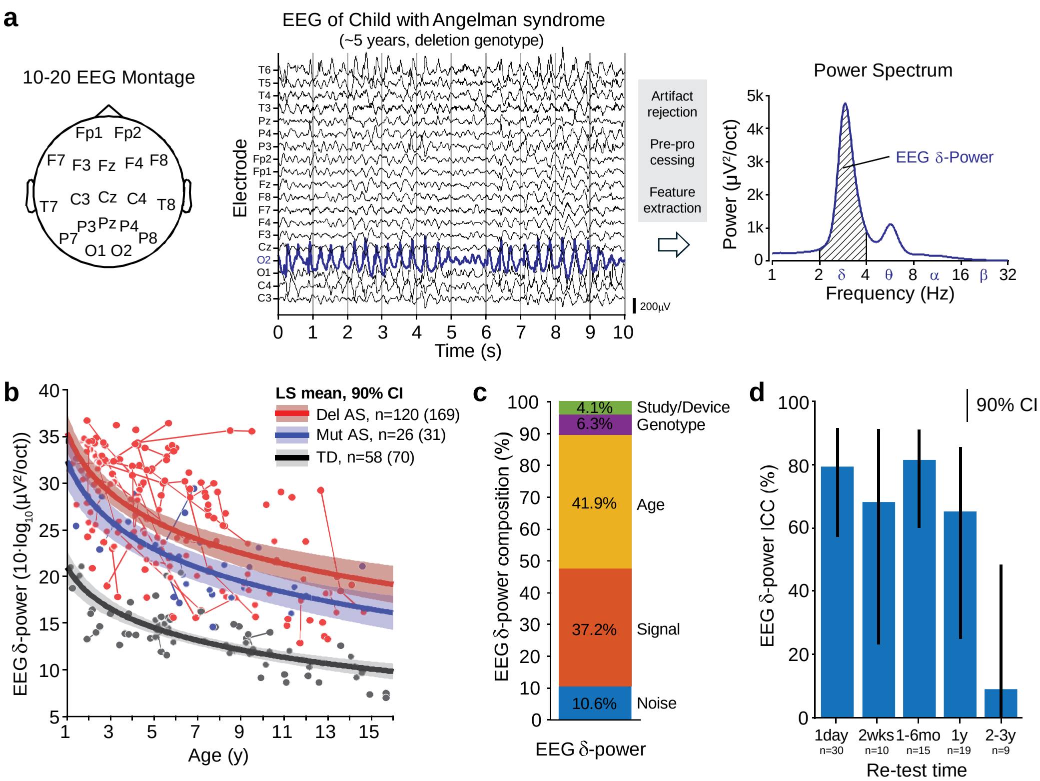

Figure S2. AS EEG natural history data. a*, EEG montage, 10 s EEG data example and associated power spectrum for a 5 year old child with Del AS.* ***b****, Developmental trajectory of EEG δ-power derived from NH data (see Online Methods).* ***c****, Composition of EEG δ-power per AS NH model. "Study/Device", "Age" and "Genotype" are fixed effects of the model, respectively. "Noise" is derived from the test-retest reliability estimate for EEG δ-power (Table S6) and "Signal" is the fraction of the EEG power presumably representing relevant information on the individual.* ***d****, Estimates of test-retest reliability / stability quantified as ICC for different re-test intervals (see also Table S5). Error bars depict 90% confidence intervals.*

We used linear mixed-effects models to quantify the developmental trajectory of EEG endpoints (see Online Methods, Fig. S2b). See Supplementary Table 21 as a separate file containing the age to EEG δ-power relationship for Del and Mut AS. Table S4 and Fig. S2c provide an overview of the different signal components comprising EEG endpoints. Age explains approximately 40% of the variance of EEG δ-power, while the fraction that presumably represents relevant information on the individual AS phenotype is generally lower but of similar magnitude.

	δ-power	Peak δ-power	IPEG δ-power	IPEG relative δ-power	Total power	Relative δ-power
Offset	4.1	3.6	3.8	-0.6	4.9	0.5
Genotype	6.3	7.1	7.4	19.8	5.7	7.3
Age	41.9	40.9	44.3	42.7	41.9	23.5
Signal	37.1	36.8	34.1	28.9	34.6	46.3
Noise	10.5	11.3	11.1	11.0	12.9	17.7

Table S4. Composition of EEG endpoints per AS NH models. *"Study/Device", "Age" and "Genotype" are the co-variates of the model, respectively (see Online Methods). "Noise" is derived from the test-retest reliability estimates of EEG endpoints (Table S6) and "Signal" is the fraction of the EEG power presumably representing relevant information on an individual.*

We used intra-class correlation (ICC) coefficients to quantify the test-retest reliability of EEG metrics[4]. The available EEG natural history data has varying lags between longitudinal visits, ranging from overnight to several years. To quantify test-retest reliability, retest intervals should be short. At the same time, we aimed at including as much data as possible. We therefore computed the ICC for EEG δ-power for different lags between visits (Table S5, Fig. S2d). For intervals of up to 6 months, the values were between 0.68 and 0.81 with a weighted average of 0.78 (Table S6), which is considered excellent[4] and was used for further analyses.

TRT Interval	ICC	ICC 5%	ICC 95%	N
Overnight	0.79	0.57	0.92	23
2 weeks	0.68	0.23	0.91	10
1 to 6 months	0.81	0.60	0.91	12
0.5 to 1.5 years	0.65	0.25	0.85	18
1.5 to 3.5 years	0.09	0.00	0.48	9

Table S5. Test-retest (TRT) reliability of EEG δ-power quantified as intra-class correlation coefficient (ICC) for different retest intervals*. 5% and 95% represent the lower and upper end of 90% confidence intervals. N is the number of data-pairs used to derive the ICC.*

We derived ICCs also for other EEG endpoints (see Online Methods), which yield similar ICC results, but were numerically smaller than EEG δ-power (Table S6). Overall, the test-retest reliabilities for EEG endpoints in AS were good to excellent.

EEG endpoint	ICC	ICC 5%	ICC 95%
δ-power	0.78	0.51	0.91
Peak δ-power	0.76	0.44	0.93
IPEG δ-power	0.75	0.51	0.91
IPEG relative δ-power	0.72	0.46	0.86
Total power	0.73	0.44	0.89
Relative δ-power	0.72	0.41	0.91

Table S6. TRT reliability of EEG δ-power quantified as ICC for retest intervals between 1 day and ½ year for different EEG endpoints. *5% and 95% represent the lower and upper end of 90% confidence intervals. The number of data-pairs used to derive the ICCs was n=45.*

7. Clinical scales NH data: dev. trajectories and analytical properties

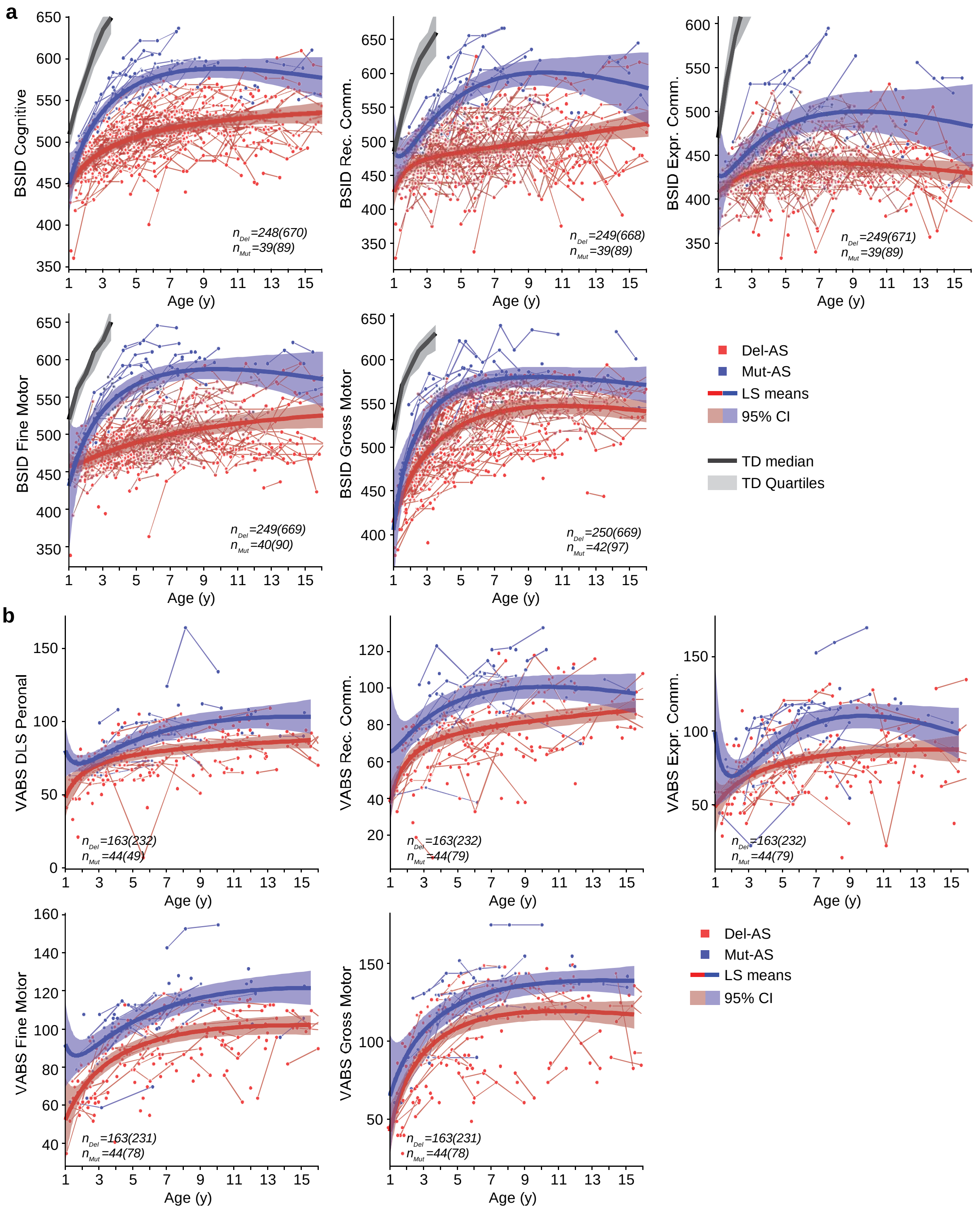

***Figure S3. BSID and VABS developmental trajectories derived from AS NH data. a**, LS means BSID-III growth score values (GSVs). TD=Typically Development (median scores and quartiles from BSID-III normative sample according to the manual). Bands show bootstrapped 95% confidence intervals. **b**, LS means VABS-3 GSV. Bands show bootstrapped 95% confidence intervals.*

We used linear mixed-effects models to quantify the developmental trajectory of clinical scores (see Online Methods, Fig. S3). See Supplementary Table 22 as a separate file containing look-up tables containing the age – clinical scales relationships for Del and Mut AS. For some scales (DLS personal, Expressive Communication and Fine Motor scales), the cubic fit with the limited number of datapoints for Mut AS led to strong transients for the youngest ages. These transients likely do not reflect the true developmental trajectories, and more work is needed to understand the youngest (and older) age range in AS. For this publication, where the youngest individual with Mut AS was older than 2.5 years at baseline, and therefore, the models should be fit for purpose.

Table S7 details the components comprising clinical scores according to the models. Furthermore, we quantified the test-retest reliability as ICC[4]. The retest intervals ranged between 1 month and 14 months and were mostly around 1 year, which is too long for a proper retest interval. The values should therefore be considered as lower bounds. Values ranged between 0.6 and 0.75, which is generally considered to be good[4].

Scale		ICC	5%	95%	n_{ICC}	P_{Covar} (%)	P_{Signal} (%)	P_{Noise} (%)
BSID	Cognitive	0.75	0.67	0.81	153	59.8	30.2	10.0
	Rec Comm	0.68	0.59	0.75	151	45.6	36.8	17.6
	Expr Comm	0.67	0.54	0.76	153	35.5	42.9	21.5
	Fine Motor	0.75	0.66	0.81	151	50.4	37.2	12.5
	Gross Motor	0.80	0.74	0.85	151	57.6	33.8	8.6
VABS	DLS Personal	0.81	0.75	0.89	44	42.1	47.1	10.8
	Rec Comm	0.75	0.61	0.85	44	41.7	43.7	14.6
	Expr Comm	0.60	0.42	0.76	44	32.1	40.7	27.1
	Fine Motor	0.72	0.56	0.83	43	55.2	32.4	12.4
	Gross Motor	0.79	0.57	0.91	43	48.3	40.9	10.8

Table S7. Analytical properties of clinical scales derived from AS NH data. *ICC: intra-class correlation coefficient (retest intervals, 1-14 months, mostly around 1 year) with 5% and 95% representing the lower and upper end of 90% confidence intervals and n_{ICC} the number of pairs of datapoints to derive the ICCs. P_{Covar}, P_{Signal} and P_{Noise} are the fraction of the variance (%) of the clinical scale data according to the natural history model (see Online Methods; Covar: covariates of the model including dataset, age and genotype; noise is estimated using the ICCs).*

8. Estimates of clinically important group differences

In the main text we report changes in clinical scores and compare those to the NH-reference. Besides problems associated with an external control arm not accounting for effects of an investigational clinical trial (see Discussion), the question remains whether a change of a clinical score is clinically meaningful[5,6]. Since there is no published information on what changes on BSID and VABS would be clinically meaningful in AS, we derived distribution-based estimates from the NH data to provide context for our results. Since there are insufficient data for anchor-based methods, we focused on estimates based on the distribution and reliability of scores.

In brief, literature suggests 0.5*SD[7] and the standard error of the measurement (SEM)[8] are often in line with individual meaningfulness thresholds, i.e. clinically meaningful absolute change at the individual level. Generally, clinically important group differences, i.e. relative difference between groups that indicate a clinically relevant benefit, are lower than individual meaningfulness thresholds and some authors argue that lower *minimal* meaningful thresholds (0.2*SD)[9,10] may apply. We derived estimates for all three measures for all 10 clinical scores used in our analyses, while correcting for all covariates as described in the Online Methods (Table S8). We show 0.2*SD *minimal* clinically important group difference estimates on top of the NH reference in Figs. 3 and 4 (red dotted lines). Generally, the changes observed in the MAD and LTE are of similar magnitude as the estimated *minimal* clinically important group difference in the MAD (Fig. 3) and go beyond in the LTE (Fig. 4).

Our considerations regarding clinically important group difference have clear limitations. Generally, anchor-based methods to derive clinically meaningful group differences are preferred over distribution-based estimates. Beyond the scope of this publication, clinically meaningful changes at an individual level are of great importance and distinct to group-based comparisons.

Scale		SD	n_{Subj}	n_{Data}	SD·0.2	SEM	SD·0.5
BSID	Cognitive	24.89	792	271	4.98	12.40	12.45
	Rec Comm	38.56	792	272	7.71	21.92	19.28
	Expr Comm	28.93	792	272	5.79	16.72	14.47
	Fine Motor	31.32	792	274	6.26	15.70	15.66
	Gross Motor	28.92	792	276	5.78	13.02	14.46
VABS	DLS Personal	12.72	311	207	2.54	5.49	6.36
	Rec Comm	15.05	311	207	3.01	7.53	7.52
	Expr Comm	19.60	311	207	3.92	12.39	9.80
	Fine Motor	12.66	311	207	2.53	6.66	6.33
	Gross Motor	19.65	311	207	3.93	8.98	9.83

Table S8. Distribution based estimates of clinically important group difference. *SD: standard deviation of the clinical scale scores (age range 1-16 years) corrected for co-variates (see Online Methods). n_{Subj} and n_{Data} are the number of participants and number of datapoints used to derive the SD. SD*0.2 is the estimate of a minimal clinically important group difference. SD*0.5 is often used as an individual meaningfulness threshold. SEM: Standard error or the measurement, i.e. sqrt(1-ICC)*SD.*

9. Additional exploratory clinical scales

In terms of exploratory clinical scales, the focus of this publication is on BSID, VABS and SAS-CGI (see Introduction). The BSID and VABS are of great importance as outcome measures in AS and have substantial NH data allowing to model a NH reference, while the SAS-CGI was the main clinician-rated instrument in this study. In addition to the BSID, VABS and SAS-CGI, other exploratory clinical scales were assessed in this study. While it is beyond the scope of this publication to report on these scales in detail (none of those has sufficient NH data to derive a NH reference), we report baseline values of the FMS, ORCA, ABC-2C, SNAKE and CASS-I to further characterize the study population and to provide data on these scales to the community (Table S2). In the following we provide a brief description of each of these scales.

Functional Mobility Scale (FMS)

The FMS[11] measures the functional mobility over three distinct distances, chosen to represent mobility in the home, at school and in the wider community. For each of three distances (5m, 50m, and 500m), a rating of 1-6 is assigned depending on the assistance required. 1 = uses wheelchair, 2 = uses a walker or frame, 3 = uses crutches, 4 = uses sticks (one or two), 5 = independent on level surfaces, 6 = independent on all surfaces, “C” = crawling (for mobility at home), “N” = does not apply [e.g., child does not complete the distance (500m)]. Higher total scores indicate better functional mobility.

Schlaffragebogen für Kinder mit Neurologischen und Anderen Komplexen Erkrankungen (SNAKE)

The SNAKE questionnaire[12] is a proxy observation instrument for children and adolescents with neurological and other complex illnesses. For the SNAKE, 5 scales (SNAKE component VI) can be derived that evaluate symptoms and consequences of sleep disturbances. A total score can be calculated for each of the individual scales: Scale 1 (Disturbances going to sleep), scale 2 (Disturbances remaining asleep), scale 3 (Arousal disorders), scale 4 (Daytime sleepiness), scale 5 (Daytime behavior disorders). A higher SNAKE score on a particular scale corresponds to more problematic sleep or sleep-related behavior in the context of a normative German sample.

Observer-Reported Communication Ability (ORCA)

The ORCA instrument[13] has been specifically developed with the AS community, to measure communication abilities of most relevance to patients with AS. The measure consists of 84 total questions with 70 behavioral items within 22 concepts/functions (e.g., refuse object, make choices) that cover expressive, receptive and pragmatic areas of communication, alongside a set of 14 descriptive items that capture important information about the individual's unique ways of communicating (e.g. modalities the individual uses, their current vocabulary, and language complexity). The questionnaire is designed to be completed by the primary caregiver who is most familiar with the individual and the ways they communicate.

Aberrant Behavior Checklist Second Edition – Community Version (ABC-2C)

The ABC-2C is an updated, empirically-derived, validated 58-item caregiver completed rating scale that measures the severity of a range of maladaptive behaviors commonly observed in children, adolescents, and adults with intellectual and developmental disabilities[14]. The ABC-2C assesses symptoms across 5 domains: irritability, social withdrawal, stereotypic behavior, hyperactive/noncompliance, and inappropriate speech. Total raw score is the sum of item scores within each domain.

Caregiver-reported Angelman Syndrome Scale for Impact (CASS-I)

The CASS-I[15] is a nine domain, caregiver-rated measure, assessing the caregiver's impression of AS. This scale is the caregiver-rated companion instrument to the SAS-CGI-S (see Online Methods). The CASS-I assesses eight domains considered relevant to AS: seizures, sleep, disruptive behavior, non-verbal and verbal expressive communication, fine motor skills, gross motor skills, cognition, and self-care. The ninth domain is a score for overall severity. For each domain, caregivers rate the impact using a 5-point scale, ranging from "not at all difficult" (1) to "very difficult" (5).

10. Relationship between BSID and EEG δ-power

To provide more context for the observed treatment-related changes in EEG δ-power, we compared the changes to the relationship between EEG δ-power and clinical symptoms in AS NH data. Two prior publications report that EEG δ-power is related to symptom severity as measured by the BSID[16,17]. We chose to re-derive the relationship here for all 5 BSID scales, since prior work was restricted to the cognitive domain of the BSID and used relative δ-power[17] or did not report slopes[16], which we need for our considerations below. Furthermore, additional NH data became available since the publication of the prior reports, which increases the robustness of the estimates derived. For the VABS-3 there was not sufficient data aligned with EEG recordings available to perform such analyses (a prior report on the relationship between the VABS[16] and EEG δ-power was based on VABS-II).

For 4 of 5 BSID scales, we found a significant negative correlation with EEG δ-power, i.e. the expected directionality ($p<0.05$; Table S9). Both EEG δ-power and clinical scales were corrected for covariates before correlation analysis, avoiding trivial correlations related to both measures changing with age. Clinical assessments and EEG measurements were within ±7 days. For the clinical scales with a significant relationship, we then derived the slope between EEG δ-power and clinical scores. Furthermore, we derived estimates of correlation values corrected for the test-retest reliability of EEG and BSID ($r_{SNR\text{-}correction}$; test-retest: Tables S5, S7)[18,19]. Correlation values were small to medium according to Cohen's definition[20].

In the analysis of the MAD data, we found average changes in EEG δ-power of -2.3dB below the NH reference on day 100 (Fig. 2a; Tables S10, S11). Across different ages and cumulative doses, the model predicted changes as high as -6dB for the older age group (5-12 years) and high cumulative doses (320 mg) (Fig. 2c; Table S12). Using the NH data derived slope between EEG δ-power and BSID, we converted EEG δ-power changes observed within this study to changes in BSID scores (Table S9). The magnitude of these estimates for the largest EEG δ-power changes in the MAD is above what we estimated to be a *minimal* clinically important group difference (0.2*SD; Table S8 / last column of Table S9). In the LTE Q16w-120mg dose regimen, changes in EEG δ-power were on average -2.3 dB (Table S15). Converting these values using the NH slope reveals values smaller than the *minimal* clinically important group difference. Of note, these values are taken at the trough level.

The above consideration, using the NH relationship between EEG δ-power and BSID scores as a context for treatment-related changes in this study, has clear limitations. First, the dynamics and magnitude of the relationship may be different in a treatment context than in the natural history. Second, in the treatment context we investigate changes within treated individuals, i.e. longitudinal, while the NH relationship is cross-sectional. Furthermore, in the context of disease modifying treatments neuronal activity as measured with the EEG may fundamentally change (e.g. an alpha-peak may emerge), which may not be captured by EEG δ-power in the NH of the disorder.

BSID Scale	r	5%	95%	p	$r_{SNR\text{-}Corr}$	n	slope	-2.3dB	-4dB	-6dB	SD·0.2
Cognitive	-0.24	-0.38	-0.08	0.008	-0.31	104	-1.14	2.63	4.57	6.85	4.98
Rec Comm	-0.16	-0.31	0.00	0.050	-0.22	105	-1.18	2.72	4.73	7.10	7.71
Expr Comm	-0.05	-0.21	0.11	0.309	-0.07	106	-	-	-	-	-
Fine Motor	-0.20	-0.35	-0.04	0.021	-0.26	107	-1.18	2.72	4.72	7.09	6.26
Gross Motor	-0.30	-0.44	-0.15	0.001	-0.38	108	-1.68	3.87	6.73	10.09	5.78

Table S9. Correlation between EEG δ-power and BSID scores. *Both, EEG δ-power and BSID scores, are corrected for covariates (see Online Methods; Covariates: dataset, age, genotype). EEG and clinical data were recorded within ±7 days of each other. r is the Pearson correlation coefficient and 5% and 95%*

*the upper and lower bounds of 90% confidence intervals. p is a p-value for a one-sided test for correlation coefficients smaller than zero (hypothesis: negative correlation). To derive the degrees of freedom, we used the number of participants (n), not the number of datapoints. $r_{SNR\text{-}corr}$, are the correlation coefficients corrected for the test-retest reliability of EEG and BSID (Tables S6, S8)[18,19]. The slope is the slope of a linear model (Scale ~ 1 + EEG). -2.3dB, -4dB and -6dB show the predicted change in clinical scores for EEG δ-power changes of -2.3, -4 and -6 dB, respectively, based on slope of the NH relationship. SD*0.2 is the estimate of a minimal clinically important group difference, reproduced from Table S9. Values of slope and predicted change are not derived for BSID Expressive Communication, since the relationship had a p-value >0.05.*

11. Accounting for the effect of maturation

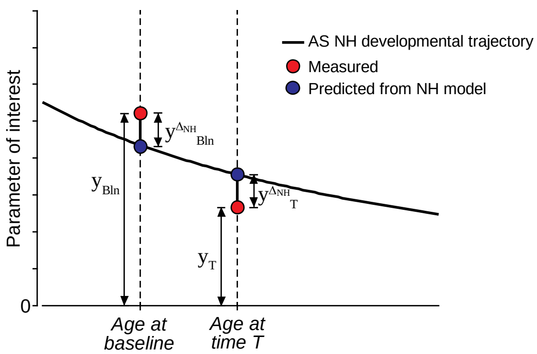

Figure S4. *Illustration of how maturation, as predicted from NH data, was accounted for in the analysis of EEG endpoints and clinical scores (see Online Methods).*

12. EEG in MAD

Visit	Estimate	SD	SE	±90% CI	p-value	dof	Decr. NH	Decr. NH HD	Decr. BL	$n_{TANGELO}$
Day 28	-0.668	2.019	0.412	0.706	0.059	24.1	65.5%		69.0%	29
Day 55	-1.421	3.280	0.511	0.860	0.004	41.2	67.4%		67.4%	46
Day 70	-0.739	3.500	0.781	1.351	0.178	20.1	52.0%	65.0%	56.0%	25
Day 100	-2.311	2.774	0.429	0.722	$1.64 \cdot 10^{-6}$	41.8	78.3%	96.4%	82.6%	46
Day 140	-2.250	2.917	0.664	1.149	0.002	19.3	79.2%	87.5%	91.7%	24
Day 224	-1.034	3.031	0.457	0.768	0.014	44.0	58.7%	78.6%	67.4%	46
	Model: $y^{\Delta_{NH}}{}_{T} \sim 1 + CumDose + y^{\Delta_{NH}}{}_{Bln}(\perp CumDose) + LogAge(\perp CumDose)$									

Table S10. MAD EEG δ-power linear model results for all timepoints. *Estimate: LS means for the intercept. SD: standard deviation. SE: standard error of the mean. ±90% CI: ± these values around the estimates provide 90% confidence intervals. P-value: p-value for one-tailed t-test for < NH reference. dof: combined degrees of freedom for TANGELO data and the NH model (is dominated by the TANGELO number of observations since smaller than the NH degrees of freedom). Decr. (Decrease): percentage of participants where the age- and baseline corrected EEG delta-band power was below the baseline (BL) or NH prediction. HD: only for participants receiving high doses (cumulative dose > 200 mg). $N_{TANGELO}$: The number of TANGELO EEG datapoints (matched baseline datapoints).*

Covariates	Estimate	SE	tStat	p-value
$1\ (Intercept)$	-2.311	0.427	-5.41	$2.79 \cdot 10^{-6}$
$CumDose$	-0.019	0.0041	-4.72	$2.59 \cdot 10^{-5}$
${y^{\Delta_{NH}}}_{Bln}(\perp CumDose)$	0.583	0.106	5.52	$1.95 \cdot 10^{-6}$
$LogAge(\perp CumDose)$	-1.484	0.547	-2.71	0.0096
Model: ${y^{\Delta_{NH}}}_T \sim 1 + CumDose + {y^{\Delta_{NH}}}_{Bln}(\perp CumDose) + LogAge(\perp CumDose)$ Number of observations: 46; Error degrees of freedom: 42; Loglikelihood: -112.11 Mean of $\log_2$(age): 2.41, corresponding to 5.3 years; Mean of cumulative dose: 220 mg				

Table S11. MAD EEG δ-power linear model fit for day 100. *Estimate: LS means estimates for all covariates. SE: standard error. tStat: t-statistics. p-value: p-values for the significance of covariates in the model, two-tailed t-test.*

Cumulative Dose	Age (y)	Estimate (dB)	±90% CI (dB)	Estimate (%)	p-value
360 mg	3.2	-3.95	1.38	40.3%	$1.53 \cdot 10^{-6}$
	5.3	-5.03	1.21	31.4%	$2.27 \cdot 10^{-12}$
	8.5	-6.03	1.36	24.9%	$7.24 \cdot 10^{-14}$
240 mg	3.2	-1.61	0.99	69.0%	$6.36 \cdot 10^{-3}$
	5.3	-2.70	0.73	53.7%	$5.56 \cdot 10^{-10}$
	8.5	-3.69	0.96	42.8%	$8.45 \cdot 10^{-11}$

Table S12. MAD EEG δ-power linear model predictions for day 100 for different ages and cumulative doses. *Estimate: prediction for intercept (model see Table S11). The estimate in dB can be converted to % using the relationship: dB = 10·$\log_{10}(EEG_{D100}/EEG_{Bln})$. Cumulative MAD doses of 240g (2 x 120 mg) and 320 mg (2 x 180 mg) and for mean log age of all participants (5.3 years), A cohorts (8.5 years) and B cohorts (3.2 years). ±90% CI: ± these values around the estimates provide 90% confidence intervals. P-value: p-value for two-tailed t-test.*

Sensitivity analyses

By design $CumDose$ and $LogAge$ are correlated. For this reason, we orthogonalized $LogAge$ with respect to $CumDose$ in the main analysis (Online Methods). However, it is not possible to fully disentangle those two. We therefore performed additional control analyses to further evaluate the robustness of our results.

No orthogonalization of $LogAge$ with respect to $CumDose$ (Table S13).

Covariates	Estimate	SE	tStat	p-value
$1\ (Intercept)$	-2.311	0.427	-5.41	$2.79 \cdot 10^{-5}$
$CumDose$	-0.016	0.0043	-3.83	$4.17 \cdot 10^{-4}$
LogAge	-1.038	0.540	-1.92	0.0612
${y^{\Delta_{NH}}}_{Bln}(\perp CumDose, \perp LogAge)$	0.584	0.106	5.52	$1.95 \cdot 10^{-6}$
Model: ${y^{\Delta_{NH}}}_T \sim 1 + CumDose + LogAge + {y^{\Delta_{NH}}}_{Bln}(\perp CumDose, \perp LogAge)$				

Table S13. MAD EEG δ-power linear model fit of for day 100 for alternative model 1. *Estimate: LS means for the covariates. SE: standard error. tStat: t-statistics. p-value: p-values for the significance of covariates in the model, two-tailed t-test.*

Swap places in terms of orthogonalization $CumDose$ <-> $LogAge$ (Table S14)

Covariates	Estimate	SE	tStat	p-value
$1\ (Intercept)$	-2.311	0.427	-5.41	$2.79 \cdot 10^{-5}$
$LogAge$	-1.852	0.521	-3.03	0.0041
$CumDose(\perp LogAge)$	-0.0092	0.0045	-2.04	0.0477
${y^{\Delta_{NH}}}_{Bln}(\perp LogAge)$	0.584	0.106	5.52	$1.95 \cdot 10^{-6}$
Model: ${y^{\Delta_{NH}}}_{T} \sim 1\ + CumDose(\perp CumDose)\ + LogAge + {y^{\Delta_{NH}}}_{Bln}(\perp LogAge)$				

Table S14. MAD EEG δ-power linear model fit of for day 100 for alternative model 2. *Estimate: LS means for the covariates. SE: standard error. tStat: t-statistics. p-value: p-values for the significance of covariates in the model, two-tailed t-test.*

We found a significant effect of $CumDose$ for all three variants, further supporting a relationship between $CumDose$ and the change in EEG δ-power.

To test if the individuals with different genotypes had a different EEG response, we compared a model including genotype as a factor (Del, Mut) to the main model for MAD day 100. Adding genotype as a categorical factor did not improve the main model (Loglikelihood ratio: 0.836, dof = 1, p = 0.361). Of note, the statistical power for the genotype comparison is small, given the low number of individuals with Mut AS (n=7). Similarly, we tested for effects of sex. Adding sex as a categorical factor revealed a trend for an improve over the main model with females showing a stronger EEG response (Loglikelihood ratio: 2.738, dof = 1, p = 0.098, coefficient, i.e. offset for male: 1.34dB). Of note, this study did not stratify for sex (see also Table 1).

Alternative EEG metrics

In the literature, different definitions of EEG δ-frequency range exist. We followed our prior work and used 2 – 4 Hz, while other authors use different definitions[21]. To allow for better comparison with other work, we performed the MAD EEG analyses for other EEG endpoints (Fig. S5a, see Online Methods for definitions). We investigated δ-power defined as 1.5 – 6 Hz, following the definition of the International Pharmaco-EEG Society (IPEG; referred to EEG IPEG δ-power) and broadband EEG power (1 – 30 Hz, referred to EEG total power). We also performed an analysis of EEG δ-peak power (spectral power at 2.8 Hz following Frohlich et al. 2019[21]; referred to as EEG δ-peak power). Some authors used relative EEG δ-power instead of absolute EEG δ-power[17,22]. We therefore performed analyses of relative power for both δ-frequency range definitions. All metrics showed qualitatively similar effects, i.e. partial normalization (decrease). Absolute power seemed to be more sensitive than relative power to rugonersen-related changes.

For analysis, we used a 10/20 EEG montage with 19 electrodes and re-referenced data to average reference (Online Methods). For some applications fewer electrodes may be preferred, e.g. bi-polar derivatives between a limited set of electrodes to reduce complexity and burden. To provide some information on other electrode configurations, we performed the main MAD EEG δ-power analysis for day 100 for all possible bi-polar derivatives of the 10/20 EEG electrodes (Fig. S5b). Effects were most pronounced for fronto-central electrode combinations and least pronounced for occipito-parietal electrode combinations.

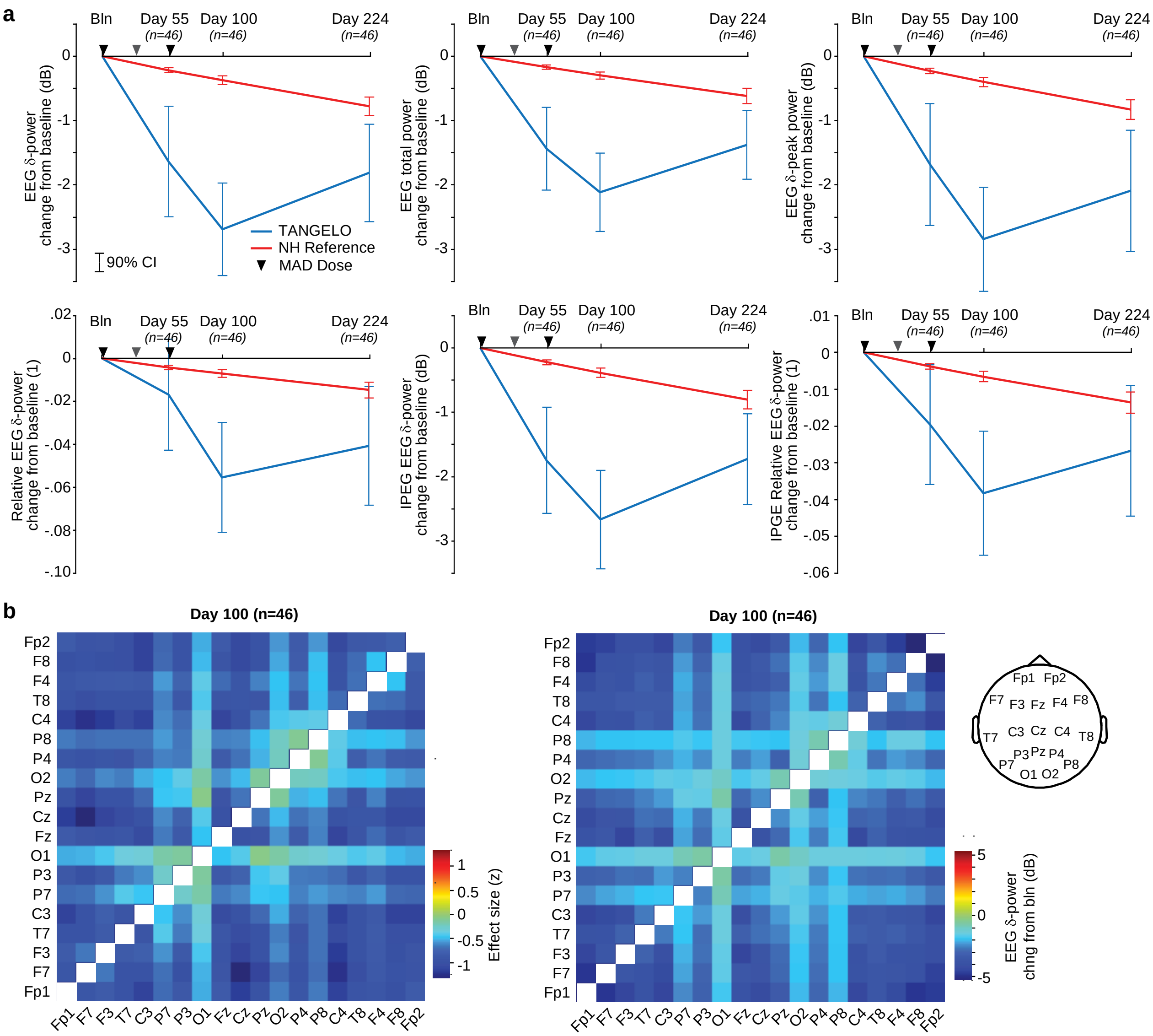

***Figure S5. Additional MAD EEG analyses. a**, LS means changes from baseline for relative EEG δ-power (2-4 Hz), total EEG power, peak EEG δ-power (2.8 Hz, as in Frohlich et al., 2019), and absolute and relative IPEG EEG δ-power (defined as 1.5 – 6 Hz). **b**, LS means EEG δ-power changes on MAD day 100 for bipolar derivatives (effect size and dB change, relative to NH reference). EEG montage.*

13. EEG in LTE

	Model: $y^{\Delta_{NH}}{}_T \sim 1 + y^{\Delta_{NH}}{}_{Bln} + LogAge$								
	Q16w-120mg								
Visit	Estimate	SD	SE	±90% CI	p-value	dof	Decrease NH	Decrease BL	$n_{TANGELO}$
LTE 1	-2.31	3.06	0.63	1.08	0.00064	23.6	82.6%	91.3%	23
LTE 2	-2.22	3.96	0.83	1.43	0.00711	22.5	73.9%	91.3%	23
LTE 3	-2.03	4.74	1.04	1.80	0.03280	20.7	81.8%	95.5%	22
LTE 4	-3.24	5.33	1.48	2.63	0.02421	13.0	75.0%	87.5%	16
LTE 5	-2.15	4.63	1.55	2.89	0.10210	8.9	58.3%	83.3%	12
LTE 6	-3.72	2.07	1.31	4.91	0.07268	2.5	100.0%	100.0%	6
LTE Avg	-2.31	3.86	0.77	1.31	0.00295	25.3	83.3%	91.7%	24
	Q24w-180mg								
Visit	Estimate	SD	SE	±90% CI	p-value	dof	Decrease NH	Decrease BL	
LTE 1	0.56	3.65	0.84	1.46	0.74	18.9	50.0%	60.0%	20
LTE 2	0.01	2.98	0.71	1.23	0.50	17.7	47.4%	73.7%	19
LTE 3	0.72	1.83	0.75	1.53	0.81	5.9	33.3%	77.8%	9
LTE Avg	0.16	2.50	0.56	0.97	0.61	20.1	38.1%	71.4%	21
	Q16w-60mg								
Visit	Estimate	SD	SE	±90% CI	p-value	dof	Decrease NH	Decrease BL	
LTE 1	-0.25	3.21	1.21	2.34	0.42	7.1	63.6%	72.7%	11
LTE 2	-1.76	3.95	1.49	2.88	0.14	7.1	72.7%	72.7%	11
LTE 3	-0.73	3.05	1.52	3.55	0.33	4.0	62.5%	75.0%	8
LTE Avg	-0.91	3.43	1.29	2.50	0.25	7.1	72.7%	72.7%	11

Table S15. LTE EEG δ-power linear model results for all timepoints. *Estimate: LS means for intercept. SD: standard deviation. SE: standard error. ±90% CI: ± these values around the estimates provide the 90% confidence intervals. P-value: p-value for one-tailed t-tests for < NH reference. dof: combined degrees of freedom for TANGELO data and the NH model (is dominated by the TANGELO number of observations since smaller than the NH degrees of freedom). LTE i corresponds to the EEG assessment following the i^{th} LTE dose, right before the $(i+1)^{th}$ dose. For the dosing regimens Q16w-120mg and Q16w-60mg, this corresponds to 16 weeks, for the dosing regimens Q24w-180mg, this corresponds to 24 weeks after the i^{th} injection. LTE Avg is a model where all available LTE visits >LTE1 are averaged for each participant. Decrease: percentage of participants where the age- and baseline corrected EEG delta-band power was below the NH prediction and below the baseline (BL).*

Covariate	Estimate	SE	tStat	p-value
$1\ (= Intercept)$	-2.314	0.723	-3.20	0.0043
$y^{\Delta_{NH}}{}_{Bln}$	0.320	0.185	1.74	0.0974
$LogAge$	-1.740	1.072	-1.62	0.1194
Model: $y^{\Delta_{NH}}{}_T \sim 1 + y^{\Delta_{NH}}{}_{Bln} + LogAge$ Number of observations: 24, Error degrees of freedom: 21; Loglikelihood: -62.80 Mean of $\log_2$(age): 2.82, corresponding to 7.06 years				

Table S16. LTE EEG δ-power linear model fit for dose regimen Q16w-120mg. *Estimate: Estimates for covariates. SE: standard error. tStat: t-statistics. p-value: p-values for the significance of covariates, two-tailed t-test.*

To test if individuals with different genotypes had a different EEG response, we compared a model including genotype as a factor (Del, Mut) with the main model for Q16w-120mg. Adding genotype as a categorical factor did not improve the main model (Loglikelihood ratio: 0.022, dof = 1, p = 0.881). Similarly, we tested for effects of sex. Adding sex as a categorical factor showed a tendency for an improvement over the main model with females showing a stronger EEG response, in line with the MAD results, see above (Loglikelihood ratio: 1.964, dof = 1, p = 0.161, coefficient, i.e. offset for male: 1.945 dB). Of note, this study did not stratify for sex (see also Table 1).

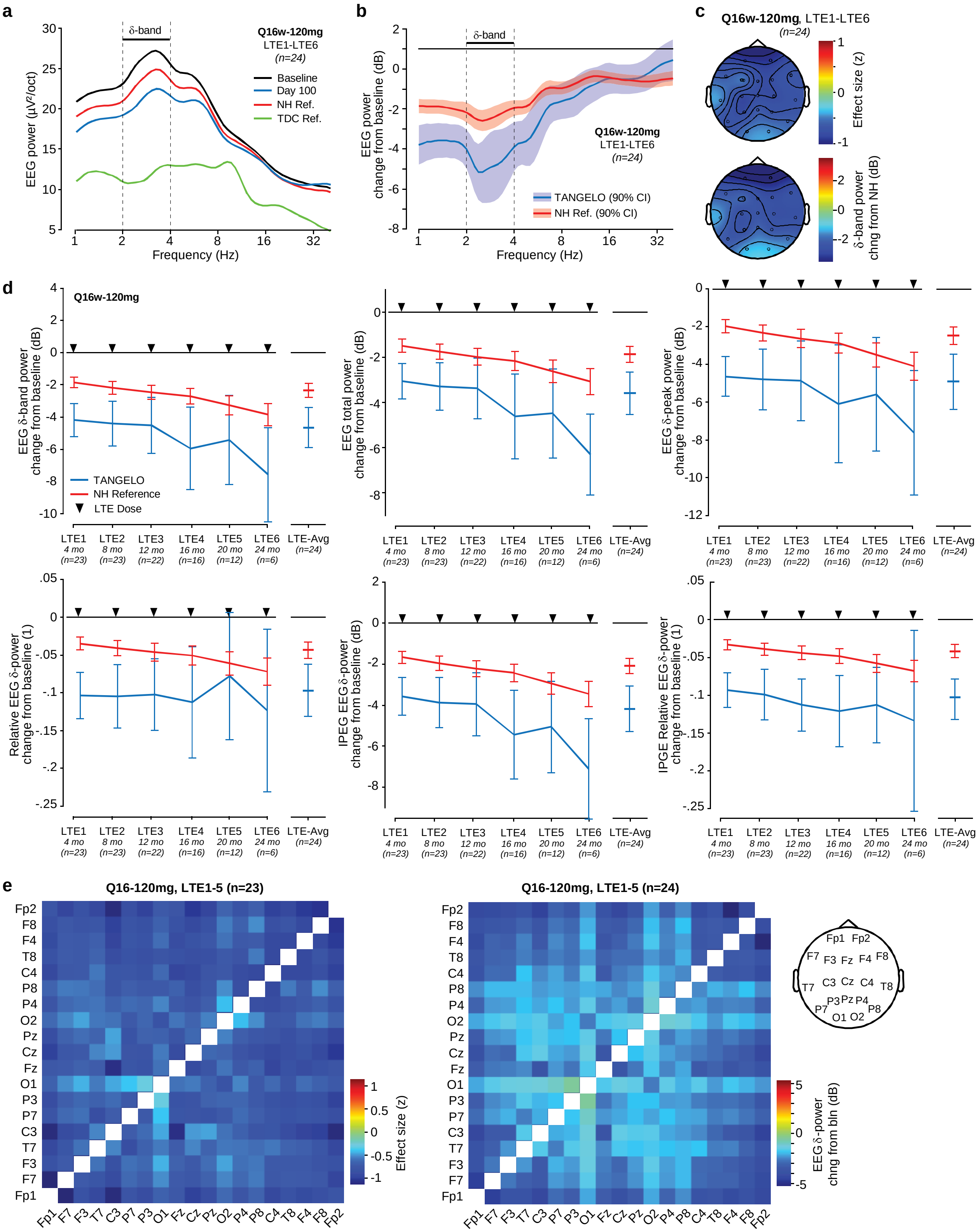

Figure S6. Additional LTE EEG analyses. a**, LS means power spectra for dose regimen Q16w-120mg (average of available LTE timepoints for each participant before averaging across participants). LTE Q16w-120mg participants at baseline (black) and day 100 (blue). Age- and genotype matched AS NH predictions (red) and age-matched predictions for typically developing children (green). **b**, LS means* power spectral change from baseline for LTE Q16w-120mg participants (blue) and age-matched NH predictions (red) on MAD day 100. ***c, Topography of LS means EEG δ-power for dose regimen Q16w-120mg averaged across all available relative to baseline and accounting for predicted changes from NH. Left: effect size, right: change in dB. ***d***, LS means for relative EEG δ-power (2-4 Hz), total EEG power, peak EEG δ-power (2.8 Hz, as defined in Frohlich et al., 2019), absolute and relative IPEG EEG δ-power (defined as 1.5 – 6 Hz). Average change from baseline in EEG δ-power for participants (blue) and age-matched NH predictions (red). LTEi corresponds to the assessment following the i^{th} LTE dose, right before the $(i+1)^{th}$ dose. LTE-Avg: average of available LTE timepoints for each participant. ***e***, EEG δ-power changes on MAD day 100 for bipolar derivatives (effect size and dB change, relative to NH reference).*

14. Clinical scales in MAD

Model: $y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln}$									
Scale		**Visit**	**Estimate**	**SD**	**SE**	**±90% CI**	**p-value**	**dof**	**$N_{TANGELO}$**
	Cognitive	Day 100	5.59	15.09	2.38	4.00	0.0119	40.33	41
	Cognitive	Day 224	3.69	14.55	2.36	3.99	0.0636	37.94	38
	Rec Comm	Day 100	4.19	20.32	3.20	5.39	0.0990	40.31	41
B	Rec Comm	Day 224	8.76	24.69	3.97	6.70	0.0168	38.64	39
S	Expr Comm	Day 100	5.29	20.23	3.12	5.25	0.0487	42.14	43
I	Expr Comm	Day 224	13.84	22.08	3.56	6.00	0.0002	38.49	39
D	Fine Motor	Day 100	3.57	21.89	3.37	5.67	0.1482	42.18	43
	Fine Motor	Day 224	8.98	23.01	3.71	6.26	0.0102	38.50	39
	Gross Motor	Day 100	-2.17	11.71	1.80	3.02	0.8832	42.41	43
	Gross Motor	Day 224	-1.12	13.22	2.12	3.57	0.7004	38.96	39
	DLS Personal	Day 100	1.25	9.19	1.44	2.43	0.1960	40.55	41
	DLS Personal	Day 224	2.43	7.31	1.24	2.10	0.0291	34.79	34
	Rec Comm	Day 100	3.04	8.62	1.36	2.29	0.0154	40.38	41
V	Rec Comm	Day 224	3.31	8.00	1.37	2.32	0.0107	34.09	34
A	Expr Comm	Day 100	2.90	8.79	1.38	2.32	0.0207	40.81	41
B	Expr Comm	Day 224	1.71	8.76	1.48	2.51	0.1294	34.79	34
S	Fine Motor	Day 100	1.11	7.95	1.25	2.11	0.1906	40.29	41
	Fine Motor	Day 224	2.00	10.65	1.83	3.09	0.1411	33.96	34
	Gross Motor	Day 100	2.41	9.49	1.51	2.54	0.0591	39.67	41
	Gross Motor	Day 224	1.41	7.93	1.33	2.25	0.1482	35.54	34

Table S17. MAD BSID and VABS scales linear model results for days 100 and 224. *Estimate: LS means for intercept. Scale: The name of the clinical scale (BSID or VABS) and the name of the domain. SD: standard deviation. SE: standard error. ±90% CI: ± these values around the estimates provide the 90% confidence intervals. P-value: p-value for one-tailed t-tests for < NH reference. dof: combined degrees of freedom for TANGELO data and the NH model (is dominated by the TANGELO number of observations since smaller than the NH degrees of freedom). $N_{TANGELO}$: The number of datapoints (matched by baseline datasets).*

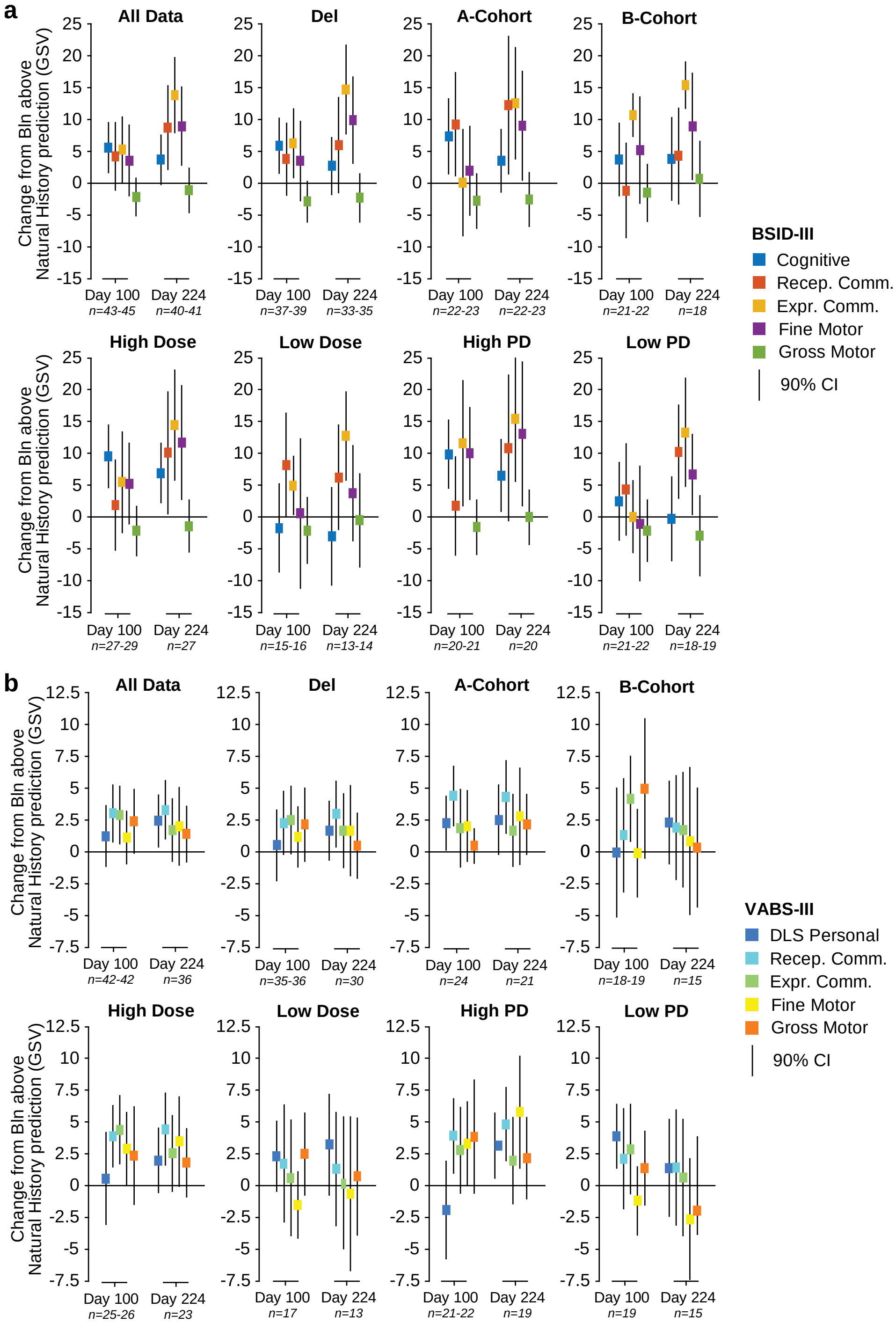

***Figure S7. Additional analyses of clinical response in the MAD. a**, BSID: LS mean changes from NH for all 5 BSID scores (color-coded) and for different sub-groups. A positive change reflects improvements. Error bars show 90% CIs. **b**, Same as (a) for VABS domains.*

Alternative models and analyses of subpopulations evaluated for MAD day 224

For the day of maximal effect in the MAD (day 224), we tested alternative models with fixed effects for age, cumulative dose, genotype, sex and EEG δ-power response (day 100, relative to NH reference) using log-likelihood ratio tests (Table S18).

Model	Scale	lrStat	pLLR	Covar	tStat	pCovar	dof
Age (A-Cohort, B-Cohort)	BSID Cognitive	0.398	0.528	3.003	0.608	0.547	37
$y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln} + CatAge$	BSID Rec Comm	0.468	0.494	-5.552	-0.660	0.513	38
B-Cohort 1-4 years at baseline	BSID Expr Comm	0.003	0.956	-0.390	-0.053	0.958	38
A-Cohort 5-12 years at baseline	BSID Fine Motor	0.007	0.935	-0.597	-0.079	0.937	38
Covariate: B-Cohort offset	BSID Gross Motor	0.666	0.414	3.377	0.789	0.435	38
	VABS DLS Pers	0.053	0.818	-0.556	-0.221	0.827	33
	VABS Rec Comm	0.018	0.893	-0.375	-0.129	0.898	33
	VABS Expr Comm	0.975	0.323	2.958	0.952	0.348	33
	VABS Fine Motor	0.111	0.739	-1.215	-0.320	0.751	33
	VABS Gross Motor	0.423	0.515	-1.672	-0.625	0.537	33
Age (continuous)	BSID Cognitive	0.023	0.878	0.513	0.147	0.884	37
$y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln} + LogAge$	BSID Rec Comm	0.902	0.342	5.378	0.919	0.364	38
Covariate: Log_2(Age) slope	BSID Expr Comm	0.095	0.758	-1.567	-0.297	0.768	38
	BSID Fine Motor	0.048	0.827	-1.097	-0.210	0.834	38
	BSID Gross Motor	0.266	0.606	-1.499	-0.498	0.622	38
	VABS DLS Pers	0.049	0.824	0.368	0.213	0.833	33
	VABS Rec Comm	0.003	0.958	0.099	0.050	0.960	33
	VABS Expr Comm	3.117	**0.077**	-3.516	-1.728	0.093	33
	VABS Fine Motor	0.090	0.764	-0.742	-0.288	0.775	33
	VABS Gross Motor	0.536	0.464	1.278	0.703	0.487	33
Cumulative Dose (High, Low)	BSID Cognitive	1.893	*0.169*	-7.169	-1.339	0.189	37
$y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln} + CatDose$	BSID Rec Comm	0.001	0.970	-0.329	-0.037	0.971	38
High: > 200 mg	BSID Expr Comm	0.239	0.625	3.650	0.471	0.640	38
Low: < 200 mg	BSID Fine Motor	0.426	0.514	-5.146	-0.630	0.532	38
Covariate: Low dose offset	BSID Gross Motor	0.069	0.793	1.170	0.252	0.802	38
	VABS DLS Pers	0.315	0.575	1.387	0.538	0.594	33
	VABS Rec Comm	0.049	0.825	-0.646	-0.212	0.833	33
	VABS Expr Comm	0.240	0.624	-1.456	-0.470	0.642	33
	VABS Fine Motor	0.575	0.448	-2.974	-0.729	0.471	33
	VABS Gross Motor	0.060	0.806	-0.657	-0.235	0.816	33
Cumulative Dose (continuos)	BSID Cognitive	0.770	0.380	0.021	0.848	0.402	37
$y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln} + CumDose$	BSID Rec Comm	0.175	0.676	-0.017	-0.403	0.689	38
Covariate: Cumulative dose slope	BSID Expr Comm	0.821	0.365	-0.031	-0.877	0.386	38
	BSID Fine Motor	1.226	0.268	0.040	1.074	0.290	38
	BSID Gross Motor	0.002	0.965	-0.001	-0.042	0.967	38
	VABS DLS Pers	0.236	0.627	-0.005	-0.466	0.644	33
	VABS Rec Comm	0.108	0.742	0.005	0.315	0.755	33
	VABS Expr Comm	0.027	0.868	0.002	0.159	0.875	33
	VABS Fine Motor	0.817	0.366	0.016	0.870	0.390	33
	VABS Gross Motor	0.378	0.539	0.007	0.590	0.559	33
Genotype (Del, Mut)	BSID Cognitive	0.542	0.462	4.440	-0.710	0.482	37
$y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln} + Genotype$	BSID Rec Comm	1.484	0.223	13.239	-1.184	0.244	38
Covariate: Mut offset	BSID Expr Comm	0.175	0.675	-3.850	0.404	0.689	38
	BSID Fine Motor	0.980	0.322	-10.267	0.959	0.344	38
	BSID Gross Motor	1.577	0.209	6.888	-1.221	0.230	38
	VABS DLS Pers	0.768	0.381	2.873	-0.843	0.405	33
	VABS Rec Comm	0.343	0.558	2.073	-0.562	0.578	33
	VABS Expr Comm	0.028	0.867	0.638	-0.160	0.874	33
	VABS Fine Motor	0.115	0.735	1.605	-0.324	0.748	33
	VABS Gross Motor	2.191	*0.139*	5.062	-1.439	0.160	33
EEG δ-power (continuous)	BSID Cognitive	2.550	*0.110*	-1.138	-1.559	0.128	35
$y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln} + EEG$	BSID Rec Comm	0.030	0.863	0.192	0.166	0.869	36
Covariate: EEG δ-power slope	BSID Expr Comm	2.011	*0.156*	1.491	1.380	0.176	36
	BSID Fine Motor	0.018	0.893	-0.141	-0.129	0.898	36
	BSID Gross Motor	0.074	0.785	-0.166	-0.262	0.795	36
	VABS DLS Pers	1.637	*0.201*	-0.404	-1.237	0.225	31
	VABS Rec Comm	0.002	0.963	0.018	0.045	0.965	31
	VABS Expr Comm	0.025	0.875	0.061	0.150	0.882	31
	VABS Fine Motor	3.849	**0.050**	-0.944	-1.928	0.063	31
	VABS Gross Motor	0.010	0.920	-0.035	-0.096	0.924	31
EEG (Strong, Weak)	BSID Cognitive	1.435	0.231	5.763	1.160	0.254	35
$y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln} + CatEEG$	BSID Rec Comm	0.010	0.919	-0.768	-0.098	0.923	36
Strong: Participants within the stronger half of	BSID Expr Comm	0.555	0.456	-5.639	-0.718	0.477	36
EEG δ-power response (i.e. more negative)	BSID Fine Motor	0.854	0.355	6.801	0.893	0.378	36
Strong: Participants within the weaker half of	BSID Gross Motor	0.489	0.484	2.904	0.674	0.505	36
EEG δ-power response (i.e. less negative) (see Fig. 2c)	VABS DLS Pers	1.169	0.280	2.720	1.041	0.306	31
Covariate: Strong EEG δ-power effect offset	VABS Rec Comm	0.248	0.619	1.428	0.476	0.637	31
	VABS Expr Comm	0.002	0.961	-0.146	-0.046	0.964	31
	VABS Fine Motor	4.983	**0.026**	8.081	2.212	0.034	31
	VABS Gross Motor	0.666	0.415	2.171	0.783	0.440	31

Sex (F, M) $y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln} + Sex$ Covariate: Male (M) offset	BSID Cognitive	2.959	**0.085**	-7.768	-1.686	0.100	37
	BSID Rec Comm	2.086	*0.149*	-11.090	-1.408	0.167	38
	BSID Expr Comm	0.418	0.518	-4.529	-0.624	0.536	38
	BSID Fine Motor	0.143	0.706	-2.730	-0.364	0.718	38
	BSID Gross Motor	0.623	0.430	-3.313	-0.763	0.450	38
	VABS DLS Pers	3.069	**0.080**	-4.154	-1.714	0.096	33
	VABS Rec Comm	0.178	0.673	-1.149	-0.405	0.688	33
	VABS Expr Comm	4.175	**0.041**	-5.745	-2.014	0.052	33
	VABS Fine Motor	1.187	0.276	-3.867	-1.052	0.301	33
	VABS Gross Motor	0.160	0.689	-1.035	-0.384	0.704	33

Table S18. Alternative models with additional fixed effects for clinical scales in the MAD on day 224. *Contribution of different fixed effects (age, dose, genotype, sex, EEG effect at MAD day 100) were tested using continuous and categorical covariates (for genotype and sex only categorical). lrStat: loglikelihood difference (degrees of freedom (dof) = 1 for all comparisons, i.e. the tested model had one more fixed effect compared to the base model). pLLR: p-value of the loglikelihood ratio test. P-values < 0.1 are shown in bold, p-values >0.1 and <0.2 in italic. In addition, p-values <0.2 are highlighted with gray background. Covar: The covariate of interest for the comparison (see also first column for more information; reference for categorical predictors is underlined). pCovar, tStat and dof provide additional information to Covar.*

15. Clinical scales in LTE

Model: $y^{\Delta_{NH}}{}_T \sim 1 + y^{\Delta_{NH}}{}_{Bln} + LogAge$									
Scale		**Visit**	**Estimate**	**SD**	**SE**	**±90% CI**	**p-value**	**dof**	**$N_{TANGELO}$**
BSID	Cognitive	LTE 1	5.56	24.00	3.42	5.73	0.0551	49.32	48
BSID	Cognitive	LTE 2	10.02	23.54	3.44	5.77	0.0027	46.86	44
BSID	Cognitive	LTE 3	15.09	24.24	3.97	6.71	0.0003	37.20	35
BSID	Cognitive	LTE Avg	13.87	22.92	3.25	5.44	$4.33 \cdot 10^{-5}$	49.86	45
BSID	Rec Comm	LTE 1	5.32	29.98	4.24	7.10	0.1076	50.05	48
BSID	Rec Comm	LTE 2	1.27	38.27	5.52	9.27	0.4099	48.00	46
BSID	Rec Comm	LTE 3	6.75	28.10	4.34	7.30	0.0637	41.95	37
BSID	Rec Comm	LTE Avg	6.28	28.72	3.88	6.50	0.0557	54.73	47
BSID	Expr Comm	LTE 1	9.82	22.44	3.15	5.28	0.0015	50.80	49
BSID	Expr Comm	LTE 2	14.03	22.66	3.16	5.30	$2.44 \cdot 10^{-5}$	51.28	47
BSID	Expr Comm	LTE 3	17.95	22.35	3.44	5.79	$2.56 \cdot 10^{-6}$	42.25	38
BSID	Expr Comm	LTE Avg	16.82	23.59	3.21	5.37	$1.34 \cdot 10^{-6}$	54.11	48
BSID	Fine Motor	LTE 1	11.19	27.14	3.83	6.42	0.0026	50.29	49
BSID	Fine Motor	LTE 2	11.13	27.97	3.97	6.65	0.0036	49.73	47
BSID	Fine Motor	LTE 3	16.72	27.63	4.39	7.40	0.0002	39.58	37
BSID	Fine Motor	LTE Avg	12.60	25.63	3.54	5.92	0.0004	52.54	47
BSID	Gross Motor	LTE 1	0.67	16.85	2.39	4.01	0.3902	49.64	48
BSID	Gross Motor	LTE 2	2.12	20.78	2.96	4.97	0.2387	49.11	47
BSID	Gross Motor	LTE 3	2.88	20.35	3.31	5.59	0.1954	37.76	36
BSID	Gross Motor	LTE Avg	2.29	19.75	2.77	4.65	0.2060	50.71	47
VABS	DLS Personal	LTE 1	3.18	8.15	1.15	1.93	0.0041	49.97	49
VABS	DLS Personal	LTE 2	3.54	8.77	1.27	2.12	0.0037	47.99	46
VABS	DLS Personal	LTE 3	4.16	7.73	1.27	2.15	0.0012	36.90	37
VABS	DLS Personal	LTE Avg	3.57	7.93	1.09	1.83	0.0010	52.83	50
VABS	Rec Comm	LTE 1	2.40	8.29	1.17	1.97	0.0231	49.98	49
VABS	Rec Comm	LTE 2	3.93	8.66	1.27	2.13	0.0017	46.52	45
VABS	Rec Comm	LTE 3	5.62	7.36	1.23	2.07	$2.67 \cdot 10^{-5}$	36.07	36
VABS	Rec Comm	LTE Avg	5.02	7.72	1.08	1.82	$1.28 \cdot 10^{-5}$	50.81	49
VABS	Expr Comm	LTE 1	1.86	10.15	1.43	2.40	0.0996	50.39	49
VABS	Expr Comm	LTE 2	3.89	12.10	1.76	2.95	0.0160	47.41	45
VABS	Expr Comm	LTE 3	5.84	13.14	2.10	3.54	0.0041	39.17	38
VABS	Expr Comm	LTE Avg	5.07	11.69	1.62	2.71	0.0014	52.14	50
VABS	Fine Motor	LTE 1	1.78	8.98	1.29	2.16	0.0862	48.69	49
VABS	Fine Motor	LTE 2	2.28	8.35	1.23	2.07	0.0358	45.80	44
VABS	Fine Motor	LTE 3	3.39	8.14	1.34	2.25	0.0077	37.11	37
VABS	Fine Motor	LTE Avg	1.95	8.56	1.21	2.02	0.0565	50.20	48
VABS	Gross Motor	LTE 1	3.53	10.02	1.42	2.38	0.0082	49.72	49
VABS	Gross Motor	LTE 2	5.06	9.66	1.43	2.41	0.0005	45.42	43
VABS	Gross Motor	LTE 3	4.56	10.67	1.79	3.02	0.0075	35.72	35
VABS	Gross Motor	LTE Avg	4.88	10.17	1.44	2.41	0.0007	49.87	47

Table S19. LTE BSID and VABS scales linear models for timepoints LTE 1-3 and the average. *Estimate: LS means for intercept. Scale: The name of the clinical scale (BSID or VABS) and the name of the domain. SD: standard deviation. SE: standard error. ±90% CI: ± these values around the estimates provide the 90% confidence intervals. P-value: p-values for one-tailed t-test for < NH reference. dof: combined degrees of freedom for TANGELO data and the NH model (is dominated by the TANGELO number of observations since smaller than the NH degrees of freedom). LTE i corresponds to the assessment following the i^{th} LTE dose, right before the $(i+1)^{th}$ dose. For the dosing regimens Q16w-120mg and Q16w-60mg, this corresponds to 16 weeks, for the dosing regimens Q24w-180mg, this corresponds to 24 weeks after the i^{th} injection. LTE Avg is a model where all LTE visits >LTE1 are averaged for each participant.*

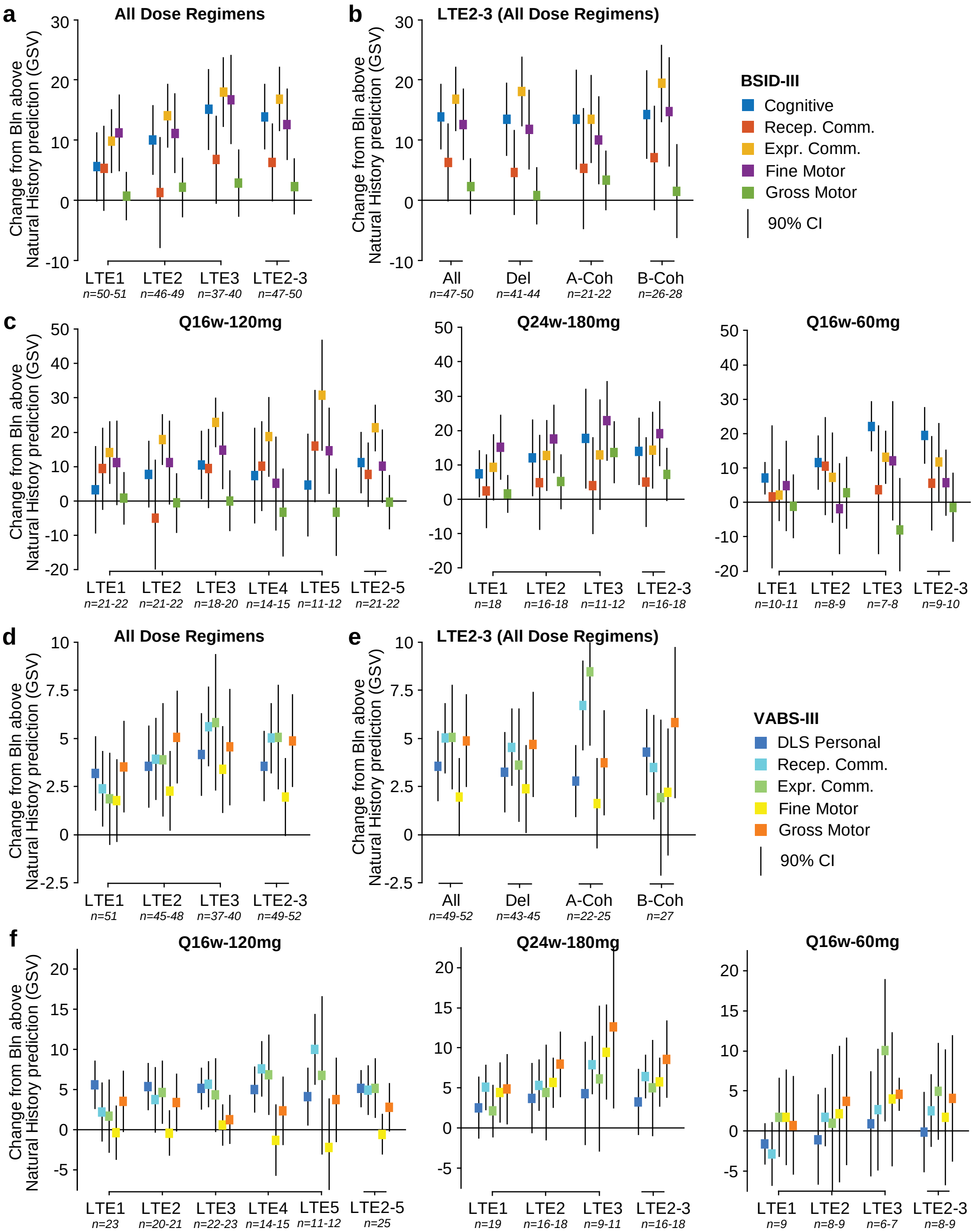

***Figure S8. Additional analyses for clinical response in the LTE. a**, BSID: LS means changes from NH reference for all 5 scales (color-coded) for all dose regimens combined. **b**, BSID: LS means changes from NH reference for all 5 domains for all dose regimens combined averaged over LTE 2 and 3 for different participant subgroups (Del: Individuals with deletion genotype only, A-Coh and B-Coh, A and B cohorts). **c,** BSID: LS means changes from NH for all LTE visits separately for each dose regimen. **d-f**, Same as (a-c) for VABS domains. Error bars show 90% CIs. Positive changes reflect improvements.*

Alternative models and analyses of subpopulations evaluated for LTE2

For the LTE2, we tested alternative models with fixed effects for age, cumulative dose, genotype, sex and EEG δ-power response (LTE2, relative to NH reference) using log-likelihood ratio tests (Table S20). LTE2 was chosen as a tradeoff between a high number of participants and late in the LTE.

Model	Scale	lrStat	pLLR	Covar	tStat	pCovar	dof
Age (<u>A-Cohort</u>, B-Cohort) $y^{\Delta NH}{}_{T} \sim 1 + y^{\Delta NH}{}_{Bln} + CatAge$ B-Cohort 1-4 years at baseline A-Cohort 5-12 years at baseline Covariate: B-Cohort offset	BSID Cognitive	0.688	0.407	5.760	0.805	0.425	43
	BSID Rec Comm	0.417	0.518	7.197	0.627	0.534	45
	BSID Expr Comm	1.080	0.299	6.269	1.013	0.317	46
	BSID Fine Motor	1.696	0.193	9.970	1.273	0.210	46
	BSID Gross Motor	0.053	0.819	-1.319	-0.222	0.825	46
	VABS DLS Pers	0.023	0.881	-0.374	-0.146	0.885	45
	VABS Rec Comm	0.005	0.944	-0.180	-0.069	0.946	44
	VABS Expr Comm	0.007	0.936	-0.278	-0.078	0.938	44
	VABS Fine Motor	0.332	0.565	1.395	0.558	0.580	43
	VABS Gross Motor	0.839	0.360	2.536	0.889	0.379	42
Age (continuous) $y^{\Delta NH}{}_{T} \sim 1 + y^{\Delta NH}{}_{Bln} + LogAge$ Covariate: Log_2(Age) slope	BSID Cognitive	0.669	0.413	-3.866	-0.794	0.432	43
	BSID Rec Comm	0.632	0.427	-6.150	-0.772	0.444	45
	BSID Expr Comm	0.370	0.543	-2.539	-0.590	0.558	46
	BSID Fine Motor	1.019	0.313	-5.360	-0.983	0.331	46
	BSID Gross Motor	0.016	0.899	0.500	0.123	0.903	46
	VABS DLS Pers	0.176	0.675	-0.715	-0.406	0.686	45
	VABS Rec Comm	0.041	0.839	0.365	0.197	0.845	44
	VABS Expr Comm	0.065	0.799	0.604	0.246	0.806	44
	VABS Fine Motor	1.881	*0.170*	-2.269	-1.340	0.187	43
	VABS Gross Motor	0.842	0.359	-1.750	-0.891	0.378	42
Dose (<u>Q16w60mg</u>, Q16w120mg, Q24w180mg) $y^{\Delta NH}{}_{T} \sim 1 + y^{\Delta NH}{}_{Bln} + CatDose$ Covariate: Q16w120mg offset	BSID Cognitive	0.399	0.819	-4.674	-0.504	0.617	42
	BSID Rec Comm	1.657	0.437	-18.640	-1.178	0.245	44
	BSID Expr Comm	1.507	0.471	8.588	0.999	0.323	45
	BSID Fine Motor	2.164	0.339	11.801	1.088	0.282	45
	BSID Gross Motor	0.619	0.734	-4.073	-0.494	0.624	45
	VABS DLS Pers	0.578	0.749	2.599	0.729	0.470	44
	VABS Rec Comm	0.800	0.670	-1.741	-0.488	0.628	43
	VABS Expr Comm	0.275	0.872	-0.471	-0.095	0.924	43
	VABS Fine Motor	4.805	**0.091**	-2.901	-0.871	0.389	42
	VABS Gross Motor	2.452	0.293	-2.394	-0.607	0.547	41
Dose (<u>Q16w60mg</u>, Q16w120mg, Q24w180mg) $y^{\Delta NH}{}_{T} \sim 1 + y^{\Delta NH}{}_{Bln} + CatDose$ Covariate: Q24w180mg offset	BSID Cognitive	0.399	0.819	-5.803	-0.584	0.562	42
	BSID Rec Comm	1.657	0.437	-9.391	-0.573	0.569	44
	BSID Expr Comm	1.507	0.471	2.009	0.226	0.822	45
	BSID Fine Motor	2.164	0.339	16.351	1.417	0.163	45
	BSID Gross Motor	0.619	0.734	0.649	0.075	0.941	45
	VABS DLS Pers	0.578	0.749	1.929	0.533	0.597	44
	VABS Rec Comm	0.800	0.670	0.644	0.176	0.861	43
	VABS Expr Comm	0.275	0.872	-2.148	-0.421	0.676	43
	VABS Fine Motor	4.805	**0.091**	2.720	0.789	0.435	42
	VABS Gross Motor	2.452	0.293	2.320	0.566	0.575	41
Genotype (<u>Del</u>, Mut) $y^{\Delta NH}{}_{T} \sim 1 + y^{\Delta NH}{}_{Bln} + Genotype$ Covariate: Mut offset	BSID Cognitive	0.324	0.569	-5.603	0.551	0.584	43
	BSID Rec Comm	0.000	1.000	0.009	0.000	1.000	45
	BSID Expr Comm	1.012	0.314	-9.208	0.980	0.332	46
	BSID Fine Motor	0.218	0.640	-5.526	0.453	0.653	46
	BSID Gross Motor	1.359	0.244	10.234	-1.137	0.261	46
	VABS DLS Pers	0.433	0.510	2.421	-0.639	0.526	45
	VABS Rec Comm	0.127	0.721	1.419	-0.346	0.731	44
	VABS Expr Comm	0.856	0.355	5.053	-0.899	0.373	44
	VABS Fine Motor	2.846	**0.092**	-5.871	1.657	0.105	43
	VABS Gross Motor	0.144	0.704	1.675	-0.367	0.715	42
EEG δ-power (continuous) $y^{\Delta NH}{}_{T} \sim 1 + y^{\Delta NH}{}_{Bln} + EEG$ Covariate: EEG δ-power slope	BSID Cognitive	0.588	0.443	-0.739	-0.742	0.462	40
	BSID Rec Comm	0.707	0.400	-1.209	-0.816	0.419	42
	BSID Expr Comm	4.351	**0.037**	-1.770	-2.065	0.045	43
	BSID Fine Motor	0.008	0.930	-0.104	-0.085	0.932	43
	BSID Gross Motor	0.082	0.774	0.242	0.277	0.783	43
	VABS DLS Pers	1.191	0.275	-0.437	-1.053	0.300	32
	VABS Rec Comm	0.609	0.435	-0.293	-0.748	0.460	31
	VABS Expr Comm	1.696	*0.193*	-0.621	-1.259	0.217	31
	VABS Fine Motor	0.390	0.532	-0.244	-0.597	0.555	30
	VABS Gross Motor	0.459	0.498	-0.311	-0.647	0.522	29
EEG (Strong, <u>Weak</u>) $y^{\Delta NH}{}_{T} \sim 1 + y^{\Delta NH}{}_{Bln} + CatEEG$ Strong: Participants within the stronger half of EEG δ-power response (i.e. more negative) Strong: Participants within the weaker half of EEG δ-power response (i.e. less negative) (see Fig. 2c) Covariate: Strong EEG δ-power effect offset	BSID Cognitive	1.768	*0.184*	9.229	1.296	0.202	40
	BSID Rec Comm	3.671	**0.055**	20.117	1.889	0.066	42
	BSID Expr Comm	5.471	**0.019**	14.769	2.330	0.025	43
	BSID Fine Motor	0.286	0.593	4.442	0.518	0.607	43
	BSID Gross Motor	0.002	0.963	0.286	0.045	0.964	43
	VABS DLS Pers	0.777	0.378	2.717	0.848	0.403	32
	VABS Rec Comm	0.161	0.689	1.126	0.383	0.704	31
	VABS Expr Comm	2.711	**0.100**	5.965	1.604	0.119	31
	VABS Fine Motor	0.735	0.391	2.563	0.822	0.418	30
	VABS Gross Motor	1.079	0.299	3.759	0.997	0.327	29

Sex (F, M) $y^{\Delta_{NH}}{}_{T} \sim 1 + y^{\Delta_{NH}}{}_{Bln} + Sex$ Covariate: Male (M) offset	BSID Cognitive	0.185	0.667	-2.827	-0.416	0.679	43
	BSID Rec Comm	3.748	**0.053**	-20.351	-1.912	0.062	45
	BSID Expr Comm	1.098	0.295	-6.486	-1.021	0.313	46
	BSID Fine Motor	0.482	0.488	-5.278	-0.674	0.504	46
	BSID Gross Motor	0.408	0.523	-3.671	-0.620	0.538	46
	VABS DLS Pers	0.165	0.684	-0.995	-0.394	0.695	45
	VABS Rec Comm	0.056	0.812	-0.590	-0.230	0.819	44
	VABS Expr Comm	0.013	0.910	-0.388	-0.110	0.913	44
	VABS Fine Motor	0.961	0.327	-2.342	-0.953	0.346	43
	VABS Gross Motor	0.878	0.349	-2.595	-0.910	0.368	42

Table S20. Alternative models with additional fixed effects for clinical scales at the LTE visit 2. *Contribution of different predictors (age, dose, genotype, sex, EEG effect at LTE visit 2) were tested using continuous and categorical fixed effects (for genotype and sex only categorical). lrStat: loglikelihood difference (degrees of freedom (dof) = 1 for all comparisons (i.e. the tested model had one more fixed effect compared to the base model). pLLR: p-value of the loglikelihood ratio test. P-values < 0.1 are shown in bold, p-values > 0.1 and < 0.2 in italic. In addition, p-values < 0.2 are highlighted with gray background. Covar: The covariate of interest for the comparison (see also first column for more information; reference for categorical predictors is underlined). pCovar, tStat and dof provide additional information to Covar.*

16. SAS-CGI-S in the LTE

For the LTE we investigated the SAS-CGI-Severity (Fig. S9), since for the SAS-CGI-Change, the reference / baseline time-point was up to several years and the past and therefore not deemed appropriate.

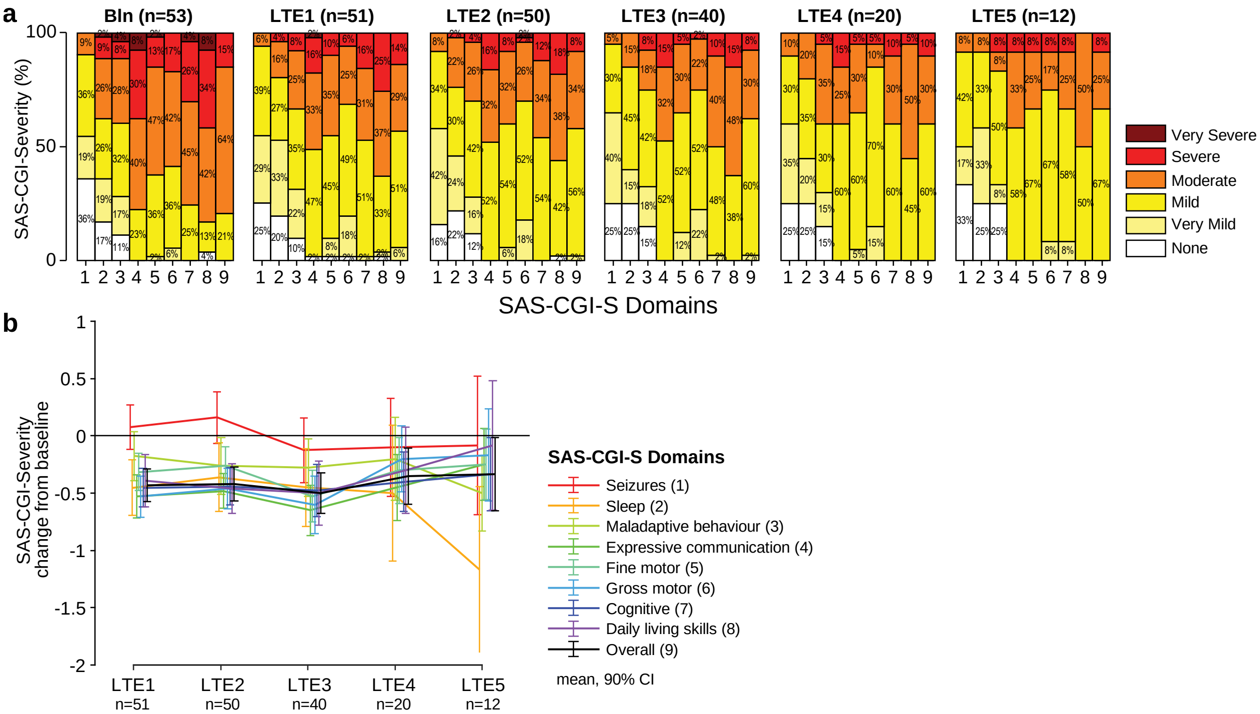

***Figure S9. SAS-CGI-Severity in LTE. a**, Frequency of SAS-CGI-Severity ratings for all 9 domains and for baseline visit and LTE1 to LTE 5 visits (none: 0, very mild: 1, … very severe: 5). **b**, Time-courses of change from baseline for all domains and the average SAS-CGI-Severity scores. Lines show means and 90% confidence intervals.*

17. Supplementary References

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